



HEART-BENCH: Do LLMs Make Persona-Consistent Decisions?

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Abstract

While LLMs have demonstrated strong task-oriented capabilities, their ability to make persona-consistent decisions grounded in long-term memory remains underexplored. We introduce HEART-BENCH, a benchmark of psychology-informed, memory-grounded persona consistency in controlled synthetic settings. HEART-BENCH defines 11 characters using Big Five-anchored profiles and value-based specifications, assigns each character 1,000 synthetic autobiographical memories, and evaluates them across 64 DIAMONDS-guided decision scenarios. Three domain experts independently compare anonymized candidates, and a senior expert selects, revises, or rewrites the final reference when needed. This process yields 673 multiple-choice questions in which a model selects the option judged most consistent with the predefined character and its history. A separate expert panel audits 176 final items, all of which receive majority support for the existing key. We evaluate 12 LLMs under a No-Retrieval baseline and three memory-augmented settings: Naive-RAG, Mem0, and PersonaDB. The strongest configuration, Gemini-3.1-Pro with Naive-RAG, reaches 63.3% accuracy, while most models remain below 40%. These results show substantial difficulty in selecting expert-adjudicated responses consistent with predefined characters and their memories.

1 Introduction

Large language models (LLMs) have achieved substantial progress in planning, reasoning, and action (Anthropic, 2026b; OpenAI, 2026b; Google DeepMind, 2026). As LLM agents are increasingly deployed in entertainment (Chen et al., 2024; Tseng et al., 2024), AI companionship (Maples et al., 2024; Phang et al., 2025),

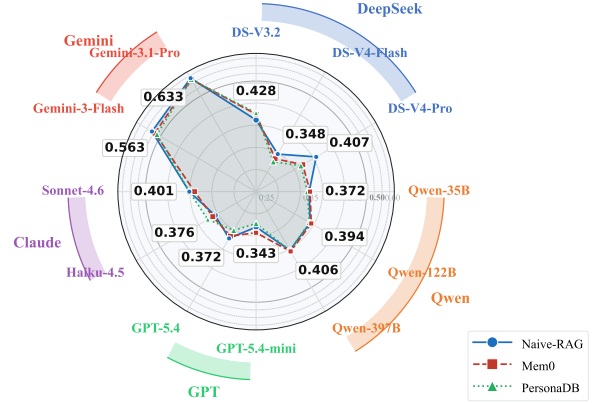


Figure 1: Accuracy for all 12 models, shown separately for the three memory-augmented settings.

and persona-cloning systems (Wei et al., 2025; Lee et al., 2026), users increasingly expect persistent and personalized behavior. These applications require more than conversational fluency and memory recall: a system should use a character’s traits, values, and prior experiences consistently when choosing how to respond. Existing evaluations, however, primarily target dialogue, explicit reasoning, or memory retrieval. This leaves a narrower but important question unresolved: *Given a predefined character, its history, and a scenario, can an LLM select a persona-consistent response?*

To address this gap, we introduce HEART-BENCH, a controlled benchmark for evaluating memory-grounded persona consistency. HEART-BENCH constructs 11 synthetic characters grounded in Big Five-anchored profiles, each associated with 1,000 structured synthetic autobiographical memories (Singer, 1995) organized into eight age-defined life-stage bins informed by developmental theories. We further design 64 decision scenarios guided by the DIAMONDS framework (Rauthmann et al., 2014). For each character-scenario pair, three domain experts independently compare candi-

Table 1: Comparison of HEART-BENCH with related benchmarks across six dimensions (see Appendix A for dimension definitions). ✓ yes; ✗ no; – partial.

Benchmark	Big Five-grounded	Age-defined life-stage coverage	Episodic memory	Scenario eval.	Expert annot.	Persona-consistent decision
CharacterEval (Tu et al., 2024)	✗	✗	✗	✓	–	✗
PersonaHub (Ge et al., 2024)	✗	✗	✗	✓	✗	✗
LoCoMo (Maharana et al., 2024)	✗	✗	✓	✓	✗	✗
CloneMem (Hu et al., 2026)	✓	–	✓	✗	✗	✗
PersonaLens (Zhao et al., 2025)	✗	✗	–	✓	–	✗
PersonaMem (Jiang et al., 2025)	✗	✗	–	✓	✗	✗
HEART-BENCH	✓	✓	✓	✓	✓	✓

date responses, and a senior expert handles every case not unanimously accepted as-is by selecting, revising, rewriting, or discarding the response. The adjudicated character response becomes the keyed reference, while responses associated with other characters in the same scenario provide distractor candidates. This process yields 673 multiple-choice questions; the memory-augmented settings test whether models can use relevant synthetic memories to identify the keyed response. Table 1 situates HEART-BENCH relative to representative persona and memory benchmarks across six design dimensions.

We evaluate 12 LLMs from the GPT, Claude, Gemini, DeepSeek, and Qwen families under a No-Retrieval baseline and three memory-augmented settings: Naive-RAG (Lewis et al., 2020), Mem0 (Chhikara et al., 2025), and PersonaDB (Sun et al., 2025). The strongest configuration, Gemini-3.1-Pro with Naive-RAG, reaches 63.3% accuracy, while most models remain below 40%. Within this controlled benchmark, the evaluated models therefore show limited ability to use encoded character and memory cues to select the expert-adjudicated response.

We summarize our contributions as follows:

- We introduce a controlled benchmark comprising 11 psychology-informed synthetic characters, 11,000 synthetic autobiographical memories, 64 decision scenarios, and 673 expert-adjudicated MCQs.
- We operationalize memory-grounded persona consistency as agreement with an expert-adjudicated reference response conditioned on a predefined character, history, and scenario, and independently audit 176 items for reference-label reliability.
- We evaluate 12 LLMs across No-Retrieval and three memory-augmented settings, quantifying their ability to use encoded

character and memory cues within this controlled synthetic setting.

2 HEART-BENCH

2.1 Overview

Figure 2 illustrates the full pipeline of HEART-BENCH. The benchmark comprises two components—an 11-character set informed by the Big Five (McCrae and John, 1992) and Schwartz’s value theory (Schwartz, 1992), with each character equipped with 1,000 synthetic long-horizon memories, and a *scenario suite* of 64 character-agnostic situations that span the eight DIAMONDS (Rauthmann et al., 2014) dimensions—Duty, Intellect, Adversity, Mat-ing, pOsitivity, Negativity, Deception, and Sociality—and are distributed across the eight age-defined life-stage bins. For each retained character–scenario pair, experts review candidate responses and adjudicate the response judged most consistent with the predefined profile and history. The resulting reference responses are assembled into MCQs in which the target character’s adjudicated response is keyed and responses associated with other characters serve as distractor candidates. After expert review, adjudication, and filtering, HEART-BENCH contains 673 evaluation instances. Across the corpus, each character’s 1,000 memories total ~1.4M tokens; the headline statistics are consolidated in Table 9 (Appendix B.7).

2.2 Character Construction

We define a synthetic character as $c = (\mathbf{p}, \{m_i\}_{i=1}^T)$, where $\mathbf{p} = (O, C, E, A, N) \in [0, 1]^5$ is a predefined Big Five vector and $\{m_i\}_{i=1}^T$ is a synthetic autobiographical history informed by research on narrative memory (Singer, 1995). We construct each character through three stages.

Stage 1: Character Profiling. We anchor each character to the Big Five factor

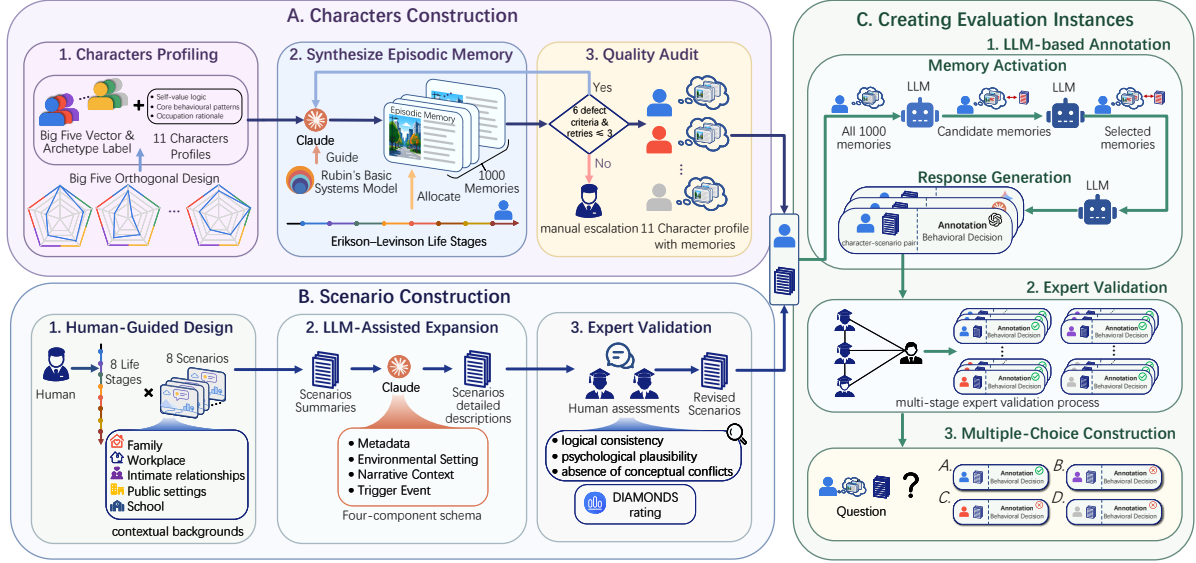


Figure 2: The pipeline has four stages: character construction, scenario construction, expert review and adjudication of reference responses, and conversion into MCQ format for model evaluation.

model (McCrae and John, 1992) rather than using a loosely defined role description. Our trait-anchored coverage design assigns one defining dimension to each profile: for every OCEAN trait, we instantiate one high ($p = 0.95$) and one low ($p = 0.10$) anchor, together with an all-neutral ($p = 0.50$) baseline, yielding 11 coherent multi-trait profiles (Table 4). For each OCEAN trait, the high and low anchors differ by 0.85 on the defining dimension, and each anchor differs from the neutral baseline by at least 0.40 on that dimension. Non-anchor dimensions vary to preserve nonuniform multi-trait character profiles. Each profile is augmented with Schwartz values derived from established trait-value correlations (Schwartz, 1992; Roccas et al., 2002), an occupation, a self-value logic, and eight behavioral patterns that guide memory generation (Appendices B.1 and B.2). The resulting profiles were reviewed by the same four-expert team that adjudicated the reference responses. This design covers both poles of every trait; differences are interpreted at the character level rather than as causal effects of an isolated trait.

Stage 2: Synthesizing Autobiographical Memories. Claude Sonnet 4.6 generates 1,000 episodic memories per character from ages 6–50. Memories are distributed across eight age-defined life-stage bins informed by Erikson (Erikson, 1963) and Levinson (Levinson et al., 1978; Levinson, 1986), with higher assigned density in adolescence and early-adult

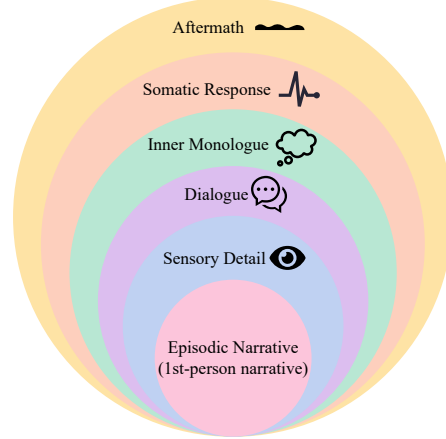


Figure 3: Five operational dimensions, inspired by Rubin’s Basic Systems Model (Rubin, 2006), used to structure synthetic episodic narratives.

bins. The 45 one-year bins also emphasize the age-30 interval. As illustrated in Figure 3, each narrative integrates five operational dimensions inspired by Rubin’s Basic Systems Model (Rubin, 2006)—sensory detail, dialogue reconstruction, inner monologue, somatic response, and aftermath—and is diversified across workplace, interpersonal, health, creative, and daily-life domains. This yields a longitudinally organized synthetic history with rich episodic content. The allocation scheme, generation protocol, and complete prompt are provided in Appendices B.3, B.4, and J.3.

Stage 3: Quality Audit. Each memory is screened against six automated defect criteria

covering perspective drift, age-inappropriate retrospection, metadata leakage, duplication, length, and schema compliance. Failures enter a targeted repair loop; memories passing these checks are then deduplicated, chronologically ordered into a 1,000-entry timeline, and sanitized with gender-neutral labels. Full criteria and repair procedures appear in Appendix B.5, with the final schema and statistics in Appendix B.7.

2.3 Scenario Construction

A scenario defines a character-agnostic decision context that serves as a shared external stimulus in our evaluation (see Appendix C.1 for a complete example). Each compatible character is placed in the same environmental and narrative context and required to respond to a decision-triggering event. During evaluation, every LLM receives the scenario, options, character ID, and occupation; memory-augmented settings additionally provide retrieved synthetic memories. The model must identify the response keyed to the target character. By holding the external situation fixed across characters, scenarios enable controlled comparison of whether LLMs can distinguish response differences encoded in the predefined profiles and synthetic histories.

Formally, a scenario is represented as $s = \langle m, \mathcal{C}_{\text{env}}, \mathcal{C}_{\text{narrative}}, e_{\text{trigger}} \rangle$, where m contains the scenario metadata, including its target life-stage bin and intensity; $\mathcal{C}_{\text{env}}$ and $\mathcal{C}_{\text{narrative}}$ specify the environmental and narrative contexts, respectively; and e_{trigger} denotes the event that prompts an immediate response.

We construct the scenario suite through a three-stage human-in-the-loop pipeline: expert blueprinting, LLM-assisted expansion, and expert review.

Stage 1: Human-Guided Blueprinting.

To ensure structural diversity, clinical experts manually draft the foundational summaries for all scenarios, using the DIAMONDS (Rauthmann et al., 2014) framework to structure coverage of situational dimensions. We design exactly 8 scenarios per life-stage bin, yielding 64 scenarios across eight bins. These summaries cover family, school, workplace, relationship, and public-space contexts. Each summary establishes a core tension while maintaining the character-agnostic constraint.

Stage 2: LLM-Assisted Expansion. We use Claude Opus 4.6 (Anthropic, 2026a) to expand the expert-drafted summaries into structured scenario descriptions based on a four-component schema (metadata, environmental setting, narrative context, and trigger event), enforcing consistent fields across age metadata and context. The full schema and a complete worked example are given in Appendix C.1.

Stage 3: Expert Review. Two doctoral-level psychology researchers review the generated scenarios for logical plausibility and assess their coverage using two theory-informed lenses: DIAMONDS (Rauthmann et al., 2014) ratings describe the eight situational dimensions at varying intensity levels, and Schwartz’s value circumplex (Schwartz, 1992) is used to check intended value-conflict alignment. The resulting rating and conflict distributions document coverage under this designed scenario set (protocol and full distributions in Appendix C.2).

2.4 Creating Evaluation Instances

Having built the character and scenario sets, we formulate each test instance as an MCQ whose keyed reference is the response judged by experts to be most consistent with the predefined target character, its synthetic history, and the scenario. This adjudicated response serves as the benchmark label.

Stage 1: Automated Annotation with LLMs. For each (character, scenario) pair, three parallel family-specific pipelines (Claude, GPT, and Gemini) retrieve relevant episodic memories and generate candidate responses conditioned on both the activated memories and the character’s full personality profile (Big Five traits, Schwartz values, and self-narrative). Using multiple model families reduces dependence on any single annotation model and provides diverse candidates for expert comparison. Full model configurations, prompts, and an annotation example are provided in Appendix D.1.

Stage 2: Expert Review and Adjudication. Three primary annotators, each with 5+ years of personality-assessment experience, independently assess pair suitability and compare anonymized candidates; a professor with expertise in psychology and computer science serves as arbitrator. Under a consensus-then-arbitration rule, a candidate is adopted directly only when all three annotators select it as *ac-*

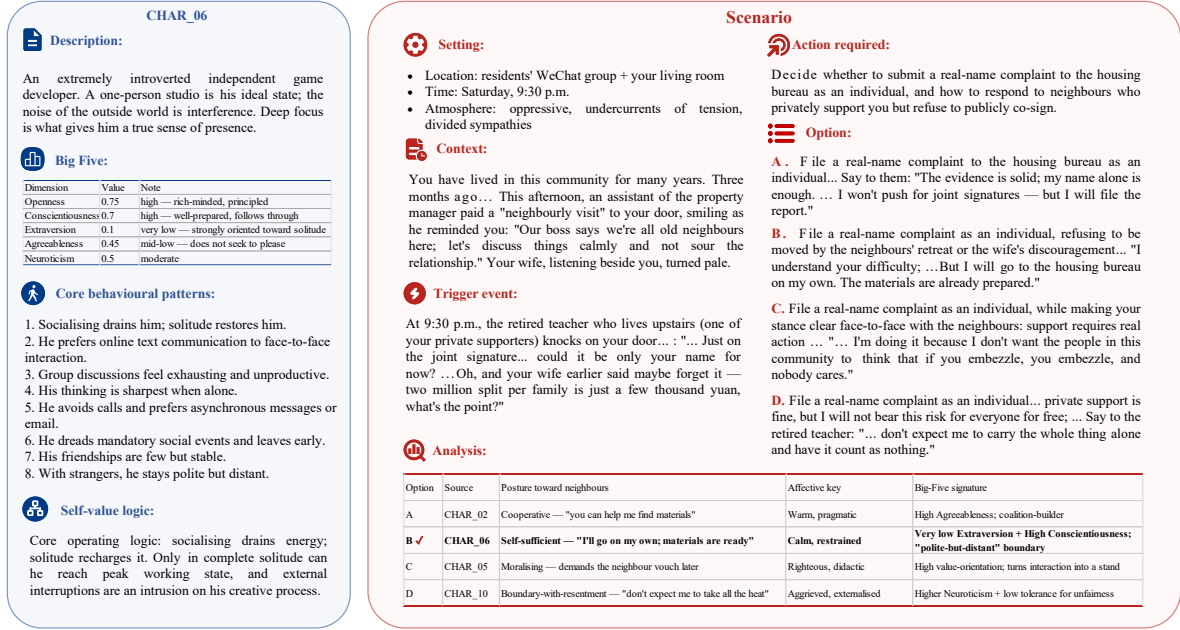


Figure 4: A HEART-BENCH instance requiring profile-response alignment beyond surface action matching; detailed analysis is in Appendix D.5.

ceptable as-is; otherwise, the arbitrator selects, edits, or rewrites the response, or discards an unsuitable pair. This process removed 31 pairs and retained 673 expert-adjudicated reference responses. Experts changed approximately 15% of the words in selected LLM candidates on average, documenting substantive human involvement in constructing the references. Full guidelines and outcome statistics are provided in Appendix D.2.

Stage 3: Test Instance Assembly. For each retained annotation, the target character's adjudicated response becomes the keyed reference option, while *Claude-Sonnet-4.6* (Anthropic, 2026c) selects three distractors from the available adjudicated responses associated with other characters in the same scenario. The selection excludes responses that are either too similar to the reference or obviously irrelevant, avoiding ambiguous or trivial questions. The four options are randomly shuffled to remove positional bias, yielding 673 MCQs. Full assembly rules are provided in Appendix D.3.

After assembly, three experts independent of the original construction team assessed a stratified random subset of 176 final MCQs. All three experts supported the existing reference option on 139 items (79.0%), while two of three supported it on the remaining 37 items (21.0%); thus, every sampled item received majority

support for the existing key. Pre-adjudication inter-rater reliability was high (Fleiss' $\kappa = 0.81$; Krippendorff's $\alpha = 0.80$). All 37 items without complete agreement entered adjudication, and the adjudicator confirmed that each item and keyed answer was usable as-is. No wording edits or keyed-answer changes were required, and the audit did not alter the evaluated dataset or primary scores. These results support reference-label reliability on the audited subset. The full protocol and results are provided in Appendix D.4.

2.5 Example of HEART-BENCH

Figure 4 illustrates profile-response alignment beyond surface action matching. CHAR_06 combines very low Extraversion (0.10), high Conscientiousness (0.70), and a "polite-but-distant" interaction pattern. Option A preserves the complaint action but adds coalition building, whereas Options C and D introduce moralizing and blame. Experts keyed Option B because its prior preparation, individual action, and concise low-contact wording best match the encoded profile. The example shows how shared actions can differ in persona-specific pragmatic framing; Appendix D.5 provides a fuller analysis.

Table 2: Accuracy on HEART-BENCH. Memory-Aug. Avg. averages the three retrieval-based settings, excluding No-Retrieval. Gemini-3.1-Pro performs best. Scenario-clustered 95% CIs are in Appendix G.

Model	No-Retrieval LLM	Naive-RAG	Mem0	PersonaDB	Memory-Aug. Avg.
<i>Open-source</i>					
DeepSeek-V3.2	0.285	0.412	0.425	0.428	0.422
DeepSeek-V4-Flash	0.278	0.348	0.336	0.328	0.337
DeepSeek-V4-Pro	0.291	0.407	0.372	0.367	0.382
Qwen3.5-35B-A3B	0.254	0.370	0.372	0.366	0.369
Qwen3.5-122B-A10B	0.264	0.391	0.394	0.389	0.391
Qwen3.5-397B-A17B	0.275	0.403	0.406	0.404	0.404
<i>Closed-source</i>					
GPT-5.4-mini	0.288	0.330	0.343	0.322	0.332
GPT-5.4	0.278	0.372	0.366	0.351	0.363
Claude-Haiku-4.5	0.245	0.357	0.363	0.376	0.365
Claude-Sonnet-4.6	0.257	0.401	0.389	0.401	0.397
Gemini-3-Flash	0.253	0.563	0.541	0.525	0.543
Gemini-3.1-Pro	0.264	0.633	0.624	0.624	0.627

3 Experiments

3.1 Experimental Setup

Model Selection. We evaluate 12 representative LLMs, including six open-source models: *DeepSeek-V3.2* (DeepSeek-AI, 2025), *DeepSeek-V4-Flash*, *DeepSeek-V4-Pro* (DeepSeek-AI, 2026), *Qwen3.5-35B-A3B*, *Qwen3.5-122B-A10B*, and *Qwen3.5-397B-A17B* (Qwen Team, 2026); and six closed-source models: *GPT-5.4-mini*, *GPT-5.4* (OpenAI, 2026a), *Claude-Haiku-4.5* (Anthropic, 2025), *Claude-Sonnet-4.6* (Anthropic, 2026c), *Gemini-3-Flash* (Google DeepMind, 2025), and *Gemini-3.1-Pro* (Google DeepMind, 2026). All models are evaluated under a unified, deterministic decoding protocol (Appendix F).

Memory-Augmented Settings. We compare three retrieval-based frameworks—Naive-RAG (Lewis et al., 2020), Mem0 (Chhikara et al., 2025), and PersonaDB (Sun et al., 2025)—against a *No-Retrieval LLM* baseline. During evaluation, all models receive the scenario, answer options, character ID, and occupation. For all three memory-augmented settings, both the memory backend and the model-facing prompt use only the de-identified `content_full` narrative; no other stored memory field enters the evaluation pipeline. All retrieval frameworks use the same embedding and auxiliary memory-management models, with retrieval budgets calibrated to provide comparable episodic content. The underlying Big Five vector, value logic, behavioral patterns, and structured psychological annotations are withheld to prevent direct reference-label leakage. Full implementation and field-isolation details are provided

in Appendix E; the model-facing evaluation template is reproduced in Appendix J.1.

Evaluation Metrics. We report *accuracy*, defined as the proportion of the 673 MCQs for which a model selects the expert-adjudicated keyed option. Only the model’s `decision_choice` field is scored; the accompanying free-text rationale fields are not compared with the reference response and do not enter the metric. This metric quantifies agreement with HEART-BENCH references. Statistical procedures are detailed in Appendix G.

3.2 Results and Discussion

Model choice dominates within the benchmark. Under Naive-RAG, the spread between the best- and worst-performing models exceeds 30% (63.3% vs. 33.0%), whereas the average gap between retrieval methods for a given model is only 1.7% and never exceeds 4%. The model- and method-axis Gini mean differences are 0.088 and 0.006, respectively, yielding a difference of 0.082 (95% CI 0.068–0.096). Gemini-3.1-Pro achieves the best accuracy (63.3%, 95% CI 59.2–67.4%) and ranks first by Memory-Augmented Avg. in all 10,000 scenario-bootstrap resamples. All non-Gemini models achieve only 33.2%–42.2% Memory-Augmented Avg., showing limited agreement with the expert-adjudicated references for synthetic characters under this setup.

Retrieval helps, with no significant differences among backends. Retrieval outperforms No-Retrieval in all 36 Holm-corrected tests, whereas no comparison among the three retrieval backends is significant. Rankings

Table 3: Memory-content ablation on the 176-item subset. Accuracy is averaged over the 12 LLMs.

Cond.	Character	source	Memory selection	Acc.
C0	Target		Target scenario	0.399
C1	Target		Other same-bin scenario	0.349
C2	Other		Target scenario	0.227
C3	Target		Random non-target	0.345

remain strongly correlated across backends ($\rho = 0.89\text{--}0.94$), and Mem0 and PersonaDB have lower point estimates than Naive-RAG for 6 and 8 of the 12 models, respectively. Their additional summarization and structural abstraction may obscure latent character cues in the original episodic memories, although the nonsignificant backend comparisons do not establish equivalence.

Memory ownership and scenario alignment are both informative. On a fixed 176-item subset, we evaluate the same 12 models under Naive-RAG with $k=30$ while varying only the supplied memory assignment: C0 uses target-scenario memories from the target character; C1 uses memories retrieved for another scenario in the same life-stage bin from the target character; C2 uses target-scenario memories from another character; and C3 uses randomly sampled eligible non-target memories from the target character. The scenario, answer options, model, decoding protocol, and number of supplied memories are held fixed; no condition exposes the latent Big Five vector, Schwartz profile, self-value logic, or behavioral templates. C0 achieves 39.9% accuracy, exceeding C1, C2, and C3 by 5.06, 17.24, and 5.43 percentage points, respectively, with all 12 models favoring C0 in each contrast. Across paired model units, all three C0 contrasts remain significant under two-sided exact sign-flip tests after Holm correction over six comparisons (adjusted $p=0.00293$ for each). Thus, scenario-matched, character-owned memories provide information beyond either arbitrary same-character memories or scenario-matched memories from another character. Full protocol and per-model results are provided in Appendix H.

Gemini’s lead persists across reference-response provenance subsets. Because Gemini-origin candidates account for 497 of the 673 adjudicated references, same-family annotation bias is a natural concern. Gemini-3.1-Pro and Gemini-3-Flash nevertheless remain

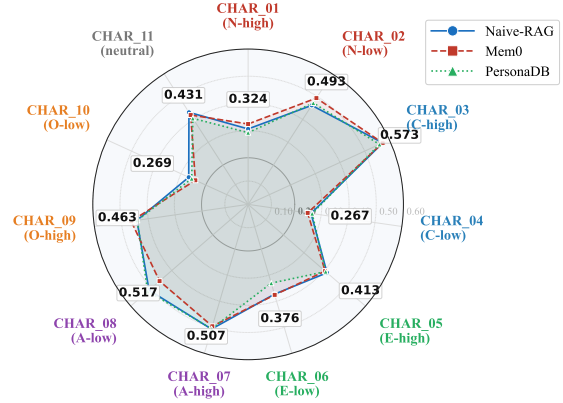


Figure 5: Per-character accuracy averaged across LLMs by memory-augmented setting.

the top two models by Memory-Augmented Avg. on the Gemini-origin (63.3% and 54.2%), Claude-origin (61.6% and 56.1%), and GPT-origin subsets (59.3% and 50.0%), with source-specific ranking correlations of $\rho = 0.666\text{--}0.809$. On non-Gemini-origin items, the Gemini family retains a 17.3-point advantage. Provenance interactions are nonsignificant in both scenario-clustered difference-in-differences (4.9 points, 95% CI $-1.1\text{--}10.8$, adjusted $p=0.3395$) and item-fixed-effect analyses (5.4 points, 95% CI $-0.6\text{--}11.5$, adjusted $p=0.2759$). These analyses do not detect a statistically significant family-specific provenance interaction, although they do not establish the absence of source bias (Appendix I.1).

Character-level difficulty and error patterns. Figure 5 provides a descriptive comparison across the synthetic characters. Several low-anchor characters are harder than their high-anchor counterparts: C-High (CHAR_03) reaches 57.3%, whereas C-Low (CHAR_04, 26.7%) and O-Low (CHAR_10, 26.9%) are the two hardest characters. One possible explanation is that the restraint, avoidance, or reduced behavioral drive encoded in these particular profiles produces less distinctive memory signals; this interpretation is exploratory rather than a general effect of low trait scores. N-High (CHAR_01, 32.4%) is a notable exception; its option distinctions may overlap with generic anxiety-like language in the synthetic responses. Errors are also concentrated: among 221 persistent shared-failure items, 84.2% have at least half of the Naive-RAG models converging on the same non-reference option. The dominant confusions involve options associated with op-

posite poles of the same Big Five dimension (e.g., C-Low→C-High, N-High→N-Low, and O-Low→O-High), showing that errors are structured rather than uniformly distributed.

4 Related Work

4.1 Persona Modeling in LLM Systems

The Big Five (OCEAN) framework (McCrae and John, 1992) is a prominent account of stable individual differences, with predictive validity across cultures (McCrae and Costa, 1997) and the lifespan (Roberts et al., 2007). Human-personality research also describes behavioral states that vary with situational demands (Fleeson, 2001), autobiographical experiences organized around self-relevant goals (Conway and Pleydell-Pearce, 2000), and developmental schemas associated with later reactions (Young et al., 2006). These theories motivate the structured synthetic artifacts in our benchmark.

Current LLM persona systems follow two broad lines. The first injects static role descriptions or trait vectors at inference time, as in Character-LLM (Shao et al., 2023), PersonaLLM (Jiang et al., 2024), PersonaAgent (Zhang et al., 2025a), and RGMem (Tian et al., 2025). The second develops long-term memory infrastructure, including MemGPT (Packer et al., 2023), HippoRAG (Gutiérrez et al., 2024), Mem0 (Chhikara et al., 2025), and A-MEM (Xu et al., 2025). Existing systems usually provide either static profiles or long-term memory stores; few benchmarks test response selection conditioned jointly on a predefined trait/value profile and its synthetic memory history.

4.2 Benchmarks for Persona- and Psychology-Informed Evaluation

Psychometric evaluation of LLMs. Prompted LLMs can simulate Big Five questionnaire responses with reasonable reliability (Serapio-García et al., 2023), and related work uses MBTI or Big Five scales to characterize model behavior. However, these approaches treat psychometric instruments as stimuli rather than testing persona-consistent decisions in controlled scenarios. They also evaluate context-free snapshots rather than response selection conditioned jointly on a profile and a structured synthetic history (McAdams,

2001; Pasupathi et al., 2007).

Persona and memory benchmarks. Conversational persona benchmarks generally test factual or stylistic consistency. Long-horizon benchmarks such as LoCoMo (Maharana et al., 2024) and LongMemEval (Wu et al., 2024) extend evaluation to multi-session dialogue but primarily assess recall. CloneMem (Hu et al., 2026), RealMem (Bian et al., 2026), and KnowMeBench (Wu et al., 2026) use richer personal evidence, yet their central targets remain recall, preference inference, or goal consistency. In contrast, HEART-BENCH evaluates memory-grounded persona-consistent response selection under controlled scenarios structured by DIAMONDS (Rauthmann et al., 2014).

Data substrates and evaluation depth. Synthetic chat logs and task-oriented interactions often lack the perceptual, affective, and introspective detail represented in autobiographical memories (Rubin, 2006), making memory-grounded character reasoning difficult to distinguish from surface pattern matching. Existing benchmarks also rarely combine content-rich episodic narratives with explicitly age-indexed synthetic histories (McAdams, 2001). HEART-BENCH addresses this gap by using dimensions inspired by Rubin’s Basic Systems Model (Rubin, 2006) to structure generated memories and organizing them into eight age-defined life-stage bins informed by developmental theories (Erikson, 1963; Levinson et al., 1978).

5 Conclusion

In this work, we presented HEART-BENCH, a controlled benchmark for memory-grounded, persona-consistent response selection in LLMs. It combines 11 predefined synthetic characters, 11,000 synthetic autobiographical memories, 64 shared decision scenarios, and 673 MCQs keyed by expert-adjudicated reference responses. Across 12 models and four evaluation settings, agreement with these references remains limited; the memory-content ablation further shows higher accuracy when supplied memories match both the target character and the scenario. These findings characterize models’ use of encoded profile and history cues within this synthetic construction. We intend HEART-BENCH as a testbed for persona consistency and memory grounding.

Limitations

The central limitation is construct scope: HEART-BENCH evaluates agreement with expert-adjudicated references for synthetic characters rather than behavior observed from real people. Expert review supports content plausibility and reference-label reliability. A single keyed option compresses potentially plural, context-dependent behavior, and the static MCQ format does not measure interactive agent behavior. Because several evaluated model families also contributed candidate responses, source-family or stylistic cues may remain despite anonymized expert adjudication and our provenance-stratified analysis.

The benchmark is also limited to 11 characters, 64 scenarios, and single-turn choices, restricting coverage of cultures, demographics, naturalistic interaction, and within-person change. Its expert-intensive construction limits scale, and the independent audit covers 176 of 673 items, supporting a dataset-level reliability estimate rather than item-wise validation of the full benchmark. Future work should incorporate human-authored candidates, distributions over plausible responses, multi-turn settings, and references derived from observed behavior to evaluate external validity.

References

Anthropic. 2025. Introducing Claude Haiku 4.5. <https://www.anthropic.com/news/claude-haiku-4-5>. Accessed: 2026-05-06.

Anthropic. 2026a. Introducing Claude Opus 4.6. <https://www.anthropic.com/news/claude-opus-4-6>. Accessed: 2026-05-06.

Anthropic. 2026b. Introducing Claude Opus 4.7. <https://www.anthropic.com/news/claude-opus-4-7>. Accessed: 2026-05-06.

Anthropic. 2026c. Introducing Claude Sonnet 4.6. <https://www.anthropic.com/news/claude-sonnet-4-6>. Accessed: 2026-05-06.

Haonan Bian, Zhiyuan Yao, Sen Hu, Zishan Xu, Shaolei Zhang, Yifu Guo, Ziliang Yang, Xueran Han, Huacan Wang, and Ronghao Chen. 2026. Realmem: Benchmarking LLMs in real-world memory-driven interaction. *arXiv preprint arXiv:2601.06966*.

Gordon H Bower. 1981. Mood and memory. *American psychologist*, 36(2):129.

Jiangjie Chen, Xintao Wang, Rui Xu, Siyu Yuan, Yikai Zhang, Wei Shi, Jian Xie, Shuang Li, Ruihan Yang, Tinghui Zhu, Aili Chen, Nianqi Li, Lida Chen, Caiyu Hu, Siye Wu, Scott Ren, Ziquan Fu, and Yanghua Xiao. 2024. From persona to personalization: A survey on role-playing language agents. *arXiv preprint arXiv:2404.18231*.

Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. 2025. Mem0: Building production-ready ai agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*.

Martin A Conway and Christopher W Pleydell-Pearce. 2000. The construction of autobiographical memories in the self-memory system. *Psychological review*, 107(2):261.

DeepSeek-AI. 2025. Deepseek-v3.2: Pushing the frontier of open large language models.

DeepSeek-AI. 2026. Deepseek-v4: Towards highly efficient million-token context intelligence.

Erik H. Erikson. 1963. *Childhood and Society*, 2nd edition. W. W. Norton & Company, New York.

William Fleeson. 2001. Toward a structure- and process-integrated view of personality: Traits as density distributions of states. *Journal of Personality and Social Psychology*, 80(6):1011–1027.

Tao Ge, Xin Chan, Xiaoyang Wang, Dian Yu, Haitao Mi, and Dong Yu. 2024. Scaling synthetic data creation with 1,000,000,000 personas. *arXiv preprint arXiv:2406.20094*.

Google DeepMind. 2025. Gemini 3 Flash model card. <https://blog.google/products-and-platforms/products/gemini/gemini-3-flash/>. Accessed: 2026-05-06.

Google DeepMind. 2026. Gemini 3.1 Pro: A smarter model for your most complex tasks. <https://blog.google/innovation-and-ai/models-and-research/gemini-models/gemini-3-1-pro/>. Accessed: 2026-05-06.

Bernal J Gutiérrez, Yiheng Shu, Yu Gu, Michihiro Yasunaga, and Yu Su. 2024. Hipporag: Neurobiologically inspired long-term memory for large language models. *Advances in neural information processing systems*, 37:59532–59569.

Sen Hu, Zhiyu Zhang, Yuxiang Wei, Xueran Han, Zhenheng Tang, Huacan Wang, and Ronghao Chen. 2026. Clonemem: Benchmarking long-term memory for ai clones. *arXiv preprint arXiv:2601.07023*.

Bowen Jiang, Zhuoqun Hao, Young-Min Cho, Bryan Li, Yuan Yuan, Sihao Chen, Lyle Ungar, Camillo J Taylor, and Dan Roth. 2025. Know me, respond to me: Benchmarking llms for dynamic user profiling and personalized responses at scale. *arXiv preprint arXiv:2504.14225*.

714	Hang Jiang, Xiajie Zhang, Xubo Cao, Cynthia Breazeal, Deb Roy, and Jad Kabbara. 2024. PersonaLLM: Investigating the ability of large language models to express personality traits. In <i>Findings of the association for computational linguistics: NAACL 2024</i> , pages 3605–3627.	766
715		767
716		768
717		769
718		
719		
720	Donggun Lee, Suyoun Lee, Hyunseung Lim, and Hwajung Hong. 2026. Creating text-based ai clones of myself: Exploring perceptions, development strategies, and challenges. <i>International Journal of Human-Computer Studies</i> , 208:103692.	770
721		771
722		772
723		773
724		774
725		
726	Daniel J. Levinson. 1986. A conception of adult development. <i>American Psychologist</i> , 41(1):3–13.	775
727		776
728		777
729	Daniel J. Levinson, Charlotte N. Darrow, Edward B. Klein, Maria H. Levinson, and Braxton McKee. 1978. <i>The Seasons of a Man’s Life</i> . Knopf, New York.	778
730		779
731		780
732		781
733	Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wentaou Yih, Tim Rocktäschel, and 1 others. 2020. Retrieval-augmented generation for knowledge-intensive nlp tasks. <i>Advances in neural information processing systems</i> , 33:9459–9474.	782
734		783
735		
736		
737		
738		
739		
740	Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. 2024. Evaluating very long-term conversational memory of llm agents. In <i>Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)</i> , pages 13851–13870.	784
741		785
742		786
743		787
744		788
745		789
746		790
747		791
748		
749		
750		
751	Bethanie Maples, Merve Cerit, Aditya Vishwanath, and Roy Pea. 2024. Loneliness and suicide mitigation for students using gpt3-enabled chatbots. <i>npj mental health research</i> , 3(1):4.	792
752		793
753		794
754		795
755		796
756		797
757		798
758		
759		
760	Dan P McAdams. 2001. The psychology of life stories. <i>Review of general psychology</i> , 5(2):100–122.	799
761		800
762		801
763		802
764		803
765		
	Robert R McCrae and Paul T Costa. 1997. Personality trait structure as a human universal. <i>American Psychologist</i> , 52(5):509–516.	804
		805
		806
	Robert R McCrae and Oliver P John. 1992. An introduction to the five-factor model and its applications. <i>Journal of personality</i> , 60(2):175–215.	807
		808
		809
		810
		811
	OpenAI. 2026a. Introducing GPT-5.4. https://openai.com/index/introducing-gpt-5-4/ . Accessed: 2026-05-06.	812
		813
		814
		815
		816
	OpenAI. 2026b. Introducing GPT-5.5. https://openai.com/index/introducing-gpt-5-5/ . Accessed: 2026-05-06.	817
		818
		819
		820
		821
	Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G Patil, Ion Stoica, and Joseph E Gonzalez. 2023. Memgpt: Towards llms as operating systems. <i>arXiv preprint arXiv:2310.08560</i> .	
	Monisha Pasupathi, Emma Mansour, and Jed R Brubaker. 2007. Developing a life story: Constructing relations between self and experience in autobiographical narratives. <i>Human development</i> , 50(2-3):85–110.	
	Jason Phang, Michael Lampe, Lama Ahmad, Sandhini Agarwal, Cathy Mengying Fang, Auren R Liu, Valdemar Danry, Eunhae Lee, Samantha WT Chan, Pat Pataranutaporn, and 1 others. 2025. Investigating affective use and emotional well-being on chatgpt. <i>arXiv preprint arXiv:2504.03888</i> .	
	Qwen Team. 2026. Qwen3.5: Towards native multimodal agents .	
	John F Rauthmann, David Gallardo-Pujol, Emmanuel M Guillaume, Emma Todd, Christopher S Nave, Robert A Sherman, Matthias Ziegler, and David C Funder. 2014. The situational eight diamonds: A taxonomy of major dimensions of situation characteristics. <i>Journal of Personality and Social Psychology</i> , 107(4):677–718.	
	Brent W Roberts, Nathan R Kuncel, Rebecca Shiner, Avshalom Caspi, and Lewis R Goldberg. 2007. The power of personality: The comparative validity of personality traits, socioeconomic status, and cognitive ability for predicting important life outcomes. <i>Perspectives on Psychological Science</i> , 2(4):313–345.	
	Sonia Roccas, Lilach Sagiv, Shalom H. Schwartz, Ariel Knafo, and Adi Anati. 2002. The Big Five personality factors and personal values. <i>Personality and Social Psychology Bulletin</i> , 28(6):789–801.	
	David C. Rubin. 2006. The basic-systems model of autobiographical memory. <i>Perspectives on Psychological Science</i> , 1(1):3–11.	
	Shalom H Schwartz. 1992. Universals in the content and structure of values: Theoretical advances and empirical tests in 20 countries. In <i>Advances in experimental social psychology</i> , volume 25, pages 1–65. Elsevier.	
	Greg Serapio-García, Mustafa Safdari, Clément Crepy, Luning Sun, Stephen Fitz, Peter Romero, Marwa Abdulhai, Aleksandra Faust, and Maja Matarić. 2023. Personality traits in large language models. <i>arXiv preprint arXiv:2307.00184</i> .	
	Yunfan Shao, Linyang Li, Junqi Dai, and Xipeng Qiu. 2023. Character-LLM: A trainable agent for role-playing. In <i>Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing</i> , pages 13153–13187.	

Jofferson A Singer. 1995. Seeing one’s self: Locating narrative memory in a framework of personality. <i>Journal of Personality</i> , 63(3):429–457.	878
Chenkai Sun, Ke Yang, Revanth Gangi Reddy, Yi Fung, Hou Pong Chan, Kevin Small, ChengXiang Zhai, and Heng Ji. 2025. Persona-db: Efficient large language model personalization for response prediction with collaborative data refinement. In <i>Proceedings of the 31st International Conference on Computational Linguistics</i> , pages 281–296.	879
Ao Tian, Yunfeng Lu, Xinxin Fan, Changhao Wang, Lanzhi Zhou, Yeyao Zhang, and Yanfang Liu. 2025. Rgmern: Renormalization group-inspired memory evolution for language agents. <i>arXiv preprint arXiv:2510.16392</i> .	880
Yu-Min Tseng, Yu-Chao Huang, Teng-Yun Hsiao, Wei-Lin Chen, Chao-Wei Huang, Yu Meng, and Yun-Nung Chen. 2024. Two tales of persona in llms: A survey of role-playing and personalization. In <i>Findings of the Association for Computational Linguistics: EMNLP 2024</i> , pages 16612–16631.	881
Quan Tu, Shilong Fan, Zihang Tian, Tianhao Shen, Shuo Shang, Xin Gao, and Rui Yan. 2024. CharacterEval: A chinese benchmark for role-playing conversational agent evaluation. In <i>Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)</i> , pages 11836–11850.	882
Jiale Wei, Xiang Ying, Tao Gao, Fangyi Bao, Felix Tao, and Jingbo Shang. 2025. Ai-native memory 2.0: Second me. <i>arXiv preprint arXiv:2503.08102</i> .	883
Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. 2024. Long-memEval: Benchmarking chat assistants on long-term interactive memory. <i>arXiv preprint arXiv:2410.10813</i> .	884
Tingyu Wu, Zhisheng Chen, Ziyang Weng, Shuhe Wang, Chenglong Li, Shuo Zhang, Sen Hu, Silin Wu, Qizhen Lan, Huacan Wang, and 1 others. 2026. Knowme-bench: Benchmarking person understanding for lifelong digital companions. <i>arXiv preprint arXiv:2601.04745</i> .	885
Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. 2025. A-mem: Agentic memory for LLM agents. <i>arXiv preprint arXiv:2502.12110</i> .	886
Jeffrey E. Young, Janet S. Klosko, and Marjorie E. Weishaar. 2006. <i>Schema Therapy: A Practitioner’s Guide</i> . Guilford Press.	887
Weizhi Zhang, Xinyang Zhang, Chenwei Zhang, Liangwei Yang, Jingbo Shang, Zhepei Wei, Henry Peng Zou, Zijie Huang, Zhengyang Wang, Yifan Gao, Xiaoman Pan, Lian Xiong, Jingguo Liu, Philip S. Yu, and Xian Li. 2025a. PersonaAgent: When large language model agents meet personalization at test time. <i>arXiv preprint arXiv:2506.06254</i> .	888
Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, Fei Huang, and Jingren Zhou. 2025b. Qwen3 embedding: Advancing text embedding and reranking through foundation models. <i>arXiv preprint arXiv:2506.05176</i> .	889
Zheng Zhao, Clara Vania, Subhradeep Kayal, Naila Khan, Shay B Cohen, and Emine Yilmaz. 2025. Personalens: A benchmark for personalization evaluation in conversational ai assistants. In <i>Findings of the Association for Computational Linguistics: ACL 2025</i> , pages 18023–18055.	890
A Benchmark Comparison Dimensions	891
Table 1 in the main text compares HEART-BENCH with related benchmarks across six dimensions. We define each dimension below. These dimensions do not exhaust all aspects of character and memory evaluation; they operationalize design properties relevant to psychology-informed, memory-grounded persona-consistent response selection in controlled scenarios.	892
• Big Five-grounded. The benchmark explicitly adopts the Big Five (OCEAN) personality framework as the basis for character design—not by merely mentioning personality, but by operationalizing it through structured trait dimensions.	893
• Age-defined life-stage coverage. The benchmark organizes synthetic memories into age-defined life-stage bins (e.g., childhood, adolescence, adulthood) informed by developmental theories (e.g., Erikson, Levinson). This criterion captures the temporal organization of synthetic memories.	894
• Episodic memory. The benchmark includes or evaluates structured synthetic autobiographical/episodic content—first-person narratives representing events, affective cues, and intended consequences for the character—rather than mere interaction logs or fact stores.	895
• Scenario eval. Evaluation requires a model to select a concrete response within	896

Table 4: The 11 character archetypes. Bolded scores mark the defining extreme dimension; CHAR_11 is the all-neutral baseline.

ID	Archetype	O	C	E	A	N	Occupation
CHAR_01	N-High	.85	.50	.30	.55	.95	Freelance content creator
CHAR_02	N-Low	.50	.55	.50	.60	.10	Community GP
CHAR_03	C-High	.45	.95	.40	.60	.30	Project manager
CHAR_04	C-Low	.80	.10	.60	.65	.45	Freelance programmer
CHAR_05	E-High	.65	.55	.95	.70	.35	Sales director
CHAR_06	E-Low	.75	.70	.10	.45	.50	Indie game developer
CHAR_07	A-High	.55	.60	.55	.95	.55	Non-profit coordinator
CHAR_08	A-Low	.65	.85	.45	.10	.35	Startup co-founder
CHAR_09	O-High	.95	.35	.65	.55	.40	Anthropology lecturer
CHAR_10	O-Low	.10	.75	.40	.55	.45	Building-materials retailer
CHAR_11	Neutral	.50	.50	.50	.50	.50	Local journalist

a structured situational context. Pure factual recall, question answering, or stylistic consistency checks do not qualify.

- **Expert annot.** The benchmark’s reference labels or quality judgments involve domain experts (e.g., psychologists, clinical researchers) as primary annotators or reviewers—not crowd workers or automated scoring alone.
- **Persona-consistent decision.** Evaluation specifically tests whether selected responses are consistent with the predefined character profile and history. Benchmarks that only test whether a model recalls facts or matches a surface stylistic pattern do not qualify.

B Character Construction and Memory Pipeline

This appendix details the four-stage pipeline used to construct the 11 character profiles and their 11,000 long-horizon memories, the unified memory schema, and the resulting dataset statistics. The pipeline structure and its connection to the benchmark are described in the main text (§2.2); here we report the implementation details and quantitative artifacts omitted for brevity.

B.1 Profile Specification

Each of the 11 character profiles is stored as a JSON object in `characters_phase11.json`. Beyond the Big Five vector and archetype label reported in Table 4, each profile contains:

- **Self-value logic** (1 sentence): the character’s core cognitive operating principle, derived from the dominant trait.

- **Core behavioral patterns** (8 items): fixed trait expressions that must appear across all generated memories.
- **Occupation rationale:** the occupation is chosen to be consistent with the dominant trait so that work-domain memories are ecologically valid.

Table 5 provides the self-value logic and three representative core behavioral patterns for each character. Full 8-pattern lists are stored in `characters_phase11.json`.

Table 5: Character specifications: self-value logic and representative behavioral patterns.

ID	Self-Value Logic	Representative Core Patterns (of 8)
CHAR_01 N-High	Pre-emptive self-attack is the primary defense against anticipated rejection.	Catastrophizing minor setbacks; ruminating on past mistakes; over-interpreting social micro-expressions.
CHAR_02 N-Low	Evidence over emotion; unverified information is not a basis for action; ambiguity is waste, not threat.	Maintaining equanimity under pressure; fact-seeking before reacting; emotional containment in conflict.
CHAR_03 C-High	Only controllable and verifiable systems are trustworthy; any deviation signals danger.	Exhaustive checklists; pre-task risk scanning; micromanaging format details even under low stakes.
CHAR_04 C-Low	The flow state is the only thing worth pursuing; plans are others' games; deadlines are external noise.	Abandoning tasks mid-way; chasing novelty over completion; losing focus outside deep-work states.
CHAR_05 E-High	A room is an opportunity; silence is waste; converting strangers into relationships is the primary metric.	Initiating social contact within minutes; thinking aloud by default; reading silence as a buying signal.
CHAR_06 E-Low	Social interaction is costly and modeled as a threat; deep work requires complete solitude.	Treating each social event as a drain needing recovery; preferring async communication; refusing open-plan settings.
CHAR_07 A-High	Conflict harms everyone; every person has goodwill if given the right frame.	Seeking the mediating position in disputes; over-attributing benign intent; postponing direct refusals.
CHAR_08 A-Low	Every interaction has a power structure; cost-benefit drives all decisions; sentiment is not data.	Mapping authority hierarchies immediately; treating conflict as information; direct assertion without packaging.
CHAR_09 O-High	Every "obvious" belief is worth questioning; fixed thinking is the primary enemy.	Deep-diving any novel idea encountered; redesigning routines into new habits; sustaining flow in creative immersion.
CHAR_10 O-Low	Proven methods are crystalized wisdom; change and novelty are costs, not opportunities.	Defending stable suppliers and routines; distrusting "innovation" discourse; valuing what works over what is new.
CHAR_11 Neutral	Balance and moderation across all dimensions; no single trait dominates decision-making.	Context-sensitive responses; moderate engagement across social and task domains; avoiding extremes.

B.2 Schwartz Value Profiles

As described in Section 2.2, each character’s Schwartz value profile is derived from its Big Five vector via the trait–value correlation structure of Roccas et al. (2002), rather than being elicited independently. Table 6 reports the three *dominant* (highest-scoring) and three *suppressed* (lowest-scoring) values for each character, with scores on the [0,1] scale; the full 19-dimensional profiles and their corresponding Big Five vectors are released with the benchmark. The derived profiles track the intended character archetypes as a sanity check: the high-Agreeableness non-profit coordinator (CHAR_07) is dominated by Benevolence values, the low-Agreeableness founder (CHAR_08) by Achievement and Power, and the low-Openness retailer (CHAR_10) by Conformity and Tradition, while the neutral baseline (CHAR_11) yields a flat profile by construction. The same four-expert team that adjudicated the reference responses (Appendix D.2) reviewed the full set of profiles for intended alignment and internal coherence with the predefined Big Five archetypes.

B.3 Memory Quota Allocation

The 45-year span (ages 6–50) is divided into 45 one-year segments. For a character that already has k memories, the algorithm computes the deficit $d = 1,000 - k$ and distributes it proportionally across segments according to a construction-time weight vector $\mathbf{w}$:

$$q_t = \left\lfloor d \cdot \frac{w_t}{\sum_j w_j} \right\rfloor + \epsilon_t \quad (1)$$

where q_t is the quota for segment t , w_t is higher for selected life-stage bins (Adolescence, Early Adult Transition, and Age-30 Transition), and $\epsilon_t \in \{0,1\}$ is a rounding correction to ensure $\sum_t q_t = d$ exactly. The weighting is inspired by findings on the temporal clustering of self-defining memories (Singer, 1995) and specifies the synthetic data distribution.

B.4 Memory Generation Protocol

Memories are generated using Claude Sonnet 4.6 via the Anthropic Messages API. Each batch covers a 2–5-year age window and targets 45–50 memories. The structured system prompt injects three constraint layers:

1. **Character constraints:** full Big Five vector, occupation, self-value logic, and all eight core behavioral patterns from Appendix B.1.
2. **Age-bin context:** target age window, life-stage-bin label, and the associated construction guidance from Table 7 (Appendix B.3).
3. **Rubin coverage:** per-dimension guidance requiring organic integration of all five dimensions without explicit section labels. Unlike a short fact record (e.g., “failed a job interview”), each `content_full` narrative is generated to represent multiple facets of an episode rather than only its propositional summary. The five dimensions are:
 - (a) *Sensory detail*—generated sights, sounds, smells, or tactile cues that situate the narrative in a specific time and place;
 - (b) *Dialogue reconstruction*—generated verbatim or near-verbatim exchanges that represent interpersonal dynamics and relational positioning beyond a short event summary;
 - (c) *Inner monologue*—generated in-the-moment appraisals designed to encode cues from the character’s trait vector $\mathbf{p}_i$ (e.g., catastrophizing for high- N);
 - (d) *Somatic response*—described physiological cues (heart rate, muscle tension, or nausea) used to encode the intended affective intensity;
 - (e) *Aftermath*—generated near-term consequences representing how the episode affects the character specification’s beliefs or coping patterns.

To prevent within-batch duplication, the prompt seeds topic diversity across five scenario domains: workplace, interpersonal, health, creative, and daily life. All batches for all 11 characters are generated in parallel, each writing to an independent file (`_charXX_bYY.txt`) for fault-tolerant incremental collection.

Structured output format. The generation prompt requests one JSON object per memory. After parsing and validation, each record is serialized for incremental storage under a unique header (`===MEM_XX_NNN===`), followed by a metadata preamble in `key: value`

Table 6: Per-character Schwartz value profiles: the three dominant (highest-scoring) and three suppressed (lowest-scoring) values, with scores on the [0,1] scale, derived from each character’s Big Five vector via the trait–value correlation structure of [Roccas et al. \(2002\)](#) and reviewed by the four-expert team (Appendix D.2). The neutral baseline CHAR_11 yields a flat profile ($s_v=0.50$ for all 19 values) by construction.

Character	Dominant values (top-3)	Suppressed values (bottom-3)
CHAR_01 (N-High)	Self-Direction-Thought (.86), Universalism-Concern (.83), Universalism-Nature (.79)	Achievement (.18), Power-Dominance (.27), Hedonism (.34)
CHAR_02 (N-Low)	Universalism-Tolerance (.69), Benevolence-Care (.68), Achievement (.66)	Security-Personal (.35), Face (.35), Power-Resources (.43)
CHAR_03 (C-High)	Conformity-Rules (.86), Benevolence-Dependability (.80), Achievement (.80)	Hedonism (.23), Stimulation (.26), Power-Resources (.33)
CHAR_04 (C-Low)	Stimulation (.92), Hedonism (.92), Universalism-Concern (.82)	Conformity-Rules (.10), Tradition (.13), Achievement (.24)
CHAR_05 (E-High)	Hedonism (.94), Stimulation (.92), Benevolence-Care (.90)	Tradition (.42), Conformity-Interpersonal (.47), Conformity-Rules (.47)
CHAR_06 (E-Low)	Self-Direction-Thought (.72), Humility (.67), Universalism-Nature (.64)	Hedonism (.18), Stimulation (.29), Benevolence-Care (.32)
CHAR_07 (A-High)	Benevolence-Care (.93), Benevolence-Dependability (.92), Universalism-Concern (.87)	Power-Dominance (.15), Power-Resources (.22), Face (.39)
CHAR_08 (A-Low)	Achievement (.94), Power-Dominance (.82), Face (.70)	Benevolence-Care (.18), Universalism-Concern (.28), Humility (.31)
CHAR_09 (O-High)	Stimulation (.95), Hedonism (.91), Universalism-Nature (.90)	Tradition (.10), Conformity-Rules (.12), Conformity-Interpersonal (.32)
CHAR_10 (O-Low)	Conformity-Rules (.94), Tradition (.93), Conformity-Interpersonal (.73)	Stimulation (.10), Self-Direction-Thought (.14), Hedonism (.15)
CHAR_11 (Neutral)	flat: $s_v = 0.50$ for all 19 values (neutral baseline, by construction)	

Table 7: Aggregate memory distribution across age-defined life-stage bins (all 11 characters, $N = 11,000$).

Stage	Count	Proportion
School age (6–12)	1,812	16.5%
Adolescence (13–18)	1,793	16.3%
Early adult trans. (19–22)	1,286	11.7%
Entering adult world (23–27)	1,489	13.5%
Age-30 transition (28–32)	1,380	12.5%
Settling down (33–40)	1,805	16.4%
Midlife transition (41–45)	795	7.2%
Entering midlife (46–50)	640	5.8%
Total	11,000	100%

format and a narrative body separated by a `--content_full--` delimiter. This storage serialization is distinct from the model-facing JSON output.

B.5 Quality Audit and Repair

The audit step screens each memory against the following six defect criteria, implemented via a combination of regex patterns and LLM-based classifiers (e.g., `audit_god_view.py`). Any failure routes the memory to the targeted repair loop.

- Perspective Drift (Omniscient Narrator).** Violating the first-person narrative constraint. Detected when the narrator uses post-hoc knowledge or future-tense markers (e.g., “years later,” “I didn’t know then”). This violates the strict “in-the-moment” constraint.
- Perspective Drift (Adult Evaluation).** Specifically flags child-age memories (≤ 12) that use adult-level cognitive appraisals or retrospective phrases (e.g., “looking back,” “I now understand”).

- Field-Length Inversion.** `context` exceeding 60 characters or `content_summary` exceeding 100 characters, indicating narrative leakage into metadata fields.

- Semantic Duplication.** Redundant life episodes within the same developmental age window. This is detected using a cosine similarity threshold of > 0.85 over TF-IDF vector representations of the narratives;

- Length Violation.** `content_full` below 2,000 characters (insufficient Rubin coverage) or above 4,500 characters (excessive verbosity).

- Schema-Vocabulary Inflation.** Use of high-register psychoanalytic terms (e.g., “abandonment schema”, “dysregulation”) in low-intensity episodes (< 0.4), indicating over-psychologization.

Repair procedure. Flagged memories enter a targeted repair loop: only the defective field(s) are regenerated using an issue-specific instruction appended to the original generation prompt. For example, a perspective-drift failure supplies the original `context` and `content_summary` and instructs the model to rewrite `content_full` from a strict first-person viewpoint. Up to three repair attempts are made per defect; memories still non-compliant after three attempts are logged for manual review. The repaired dataset is saved incrementally every five fixes to prevent data loss.

B.6 Memory Schema

Table 8 specifies all ten fields of the unified memory schema. Fields are grouped by functional role: structural (S), retrieval-index (R), and psychological annotation (P).

Table 8: Memory schema field specification. **S** = Structural (core identity and narrative fields); **R** = Retrieval-index (fields used for benchmark construction); **P** = Psychological annotation (fields encoding personality and affect). Only `content_full` supplies memory content to evaluation-time backends and prompts; all **R** and **P** fields are excluded.

Role	Field	Constraint	Description
S	id	<i>MEM_XX_NNN</i>	Unique identifier encoding character and sequence number.
S	timeline	<i>stage label (age)</i>	Life-stage tag from Table 7.
S	context	<i>15–30 chars</i>	Compact situational setting (where / when).
S	content_summary	<i>20–40 chars</i>	One-sentence event gist; used for fast retrieval pre-filtering.
S	content_full	<i>2,000–4,500 chars</i>	First-person narrative embedding all five Rubin dimensions.
R	triggers	<i>4 keywords</i>	Cue words that activate this memory during simulation.
R	relevance_tags	<i>4 tags</i>	Thematic tags linking memory to life threads and early maladaptive schema (EMS) clusters.
P	psych_conclusion	<i>1 sentence</i>	Long-term psychological interpretation of the event.
P	behavior_policy	<i>1 sentence</i>	Implicit behavioral rule reinforced by this memory (Young et al., 2006).
P	emotion_signature	<i>{primary, secondary, intensity ∈ [0, 1]}</i>	Affective signature supporting mood-congruent retrieval (Bower, 1981); intensity amplified for high-N characters.

B.7 Dataset Statistics

The headline statistics of HEART-BENCH are summarized below; per-character detail follows in Table 10.

Table 9: Statistics of HEART-BENCH.

Statistic	Value
Characters	11
Memories / character	1,000
Length / character	~1.4M tokens
Life stages	8 (ages 6–50)
Scenarios	64
MCQs	673

Table 10: Per-character dataset statistics.

ID	Archetype	Memories	Avg. Length (chars)	Avg. Intensity
CHAR_01	N-High	1,000	2856.6	0.697
CHAR_02	N-Low	1,000	3374.9	0.499
CHAR_03	C-High	1,000	3497.3	0.640
CHAR_04	C-Low	1,000	2927.0	0.652
CHAR_05	E-High	1,000	3519.4	0.659
CHAR_06	E-Low	1,000	3326.2	0.625
CHAR_07	A-High	1,000	3445.8	0.621
CHAR_08	A-Low	1,000	2939.0	0.598
CHAR_09	O-High	1,000	3485.6	0.659
CHAR_10	O-Low	1,000	3540.6	0.612
CHAR_11	Neutral	1,000	3405.8	0.561
Total / Mean	—	11,000	3307.2	0.620

Per-character statistics. Table 10 reports the finalized per-character memory counts, aver-

age `content_full` lengths, and mean encoded emotion intensities. All 11 characters reached the target of exactly 1,000 memories. The mean narrative length across the corpus is 3,307.2 characters, and the mean encoded intensity is 0.620; N-High (CHAR_01) has the highest value (0.697) and N-Low (CHAR_02) the lowest (0.499) in this synthetic corpus.

Age-defined life-stage-bin distribution.

Table 7 reports the aggregate memory distribution across age-defined stages (pooled over all 11 characters). Higher density in School age and Adolescence (combined 32.8%) and in the early-career stages (Early adult trans. + Entering adult world, 25.2%) reflects the designed quota scheme in Appendix B.3, which was informed by work on self-defining memories (Singer, 1995).

C Scenario Construction and Expert Review

This appendix documents the schema, expert review and coverage assessment, and an end-to-end example of the 64 character-agnostic scenarios constructed in §2.3.

C.1 Scenario Schema and Example

Table 11 details the four-component schema used by the LLM-assisted expansion stage (§2.3, Stage 2). The schema enforces a uniform structure across all 64 scenarios so that downstream evaluation, retrieval, and reference-response annotation can rely on stable field semantics. The system prompt that drives the expansion—including the field-level constraints, the character-agnostic principles, and the controlled vocabularies for Big Five and Schwartz keys—is reproduced in Appendix J.4.

Table 11: Four-component schema for scenario expansion with detailed descriptions.

Component	Purpose	Content Description
Metadata	Anchoring & summary	id, name, description, life stage, age, intensity level
Environmental Setting	Contextual sphere	Physical location, specific time, emotional/social atmosphere
Narrative Context	Background establishment	Detailed situational buildup, prior events, social dynamics, character relationships
Trigger Event	Decision catalyst	Specific sender and verbatim dialogue or message content that prompts an immediate response

The following example illustrates a fully expanded scenario instance under this schema. It shows how attributes such as the life-stage bin, DIAMONDS dimension, and contextual narrative are instantiated into a concrete social situation involving high-stakes decision making while maintaining the character-agnostic constraint.

Scenario Instance

```
{
  "id": "SCN_SCHOOL_AGE_3",
  "stage": "school_age",
  "age_range": "6-12 years",
  "age": 10,
  "diamonds_dimension": "Adversity",
  "name": "The Shattered Group Presentation",
  "category": "Integrity Dilemma / Responsibility Crisis",
  "intensity": "High",
  "description_for_agent": "A high-pressure adversity scenario involving a teammate's failure, loss of a core project artifact, and an imminent public presentation requiring a moral decision.",
  "setting": {
    "location": "Elementary school multipurpose auditorium backstage waiting area",
    "time": "09:45 AM",
    "atmosphere": "Noisy, tense, and suffocating"
  },
  "context_text": "Today is the school's annual \"Final Innovation Showcase Competition\". Students, homeroom teachers, the dean of studies, the principal, and a dense crowd of parents are seated in the auditorium. The air conditioning seems to be broken, making the space stiflingly hot. Your five-person group was randomly assigned by the teacher, but over the past two weeks, almost all research, writing, and rehearsal work has been completed by you alone. The only delegated task - transporting your group's \"solar system model\" built from cardboard, clay, and LED lights - was assigned to a group member who lives nearby, just two bus stops away. Backstage, harsh fluorescent lights illuminate the waiting area. You can hear the previous group presenting \"Plant Respiration\", occasionally interrupted by judges asking questions in low voices. The host has just announced: \"Next group, please get ready.\" You turn toward the entrance - the teammate responsible for the model has not yet arrived.",
  "trigger_event": {
    "sender": "Responsible group member",
```

Scenario Instance (continued)

```
"message_content": "(bursts in through the side entrance of the backstage area, sweating heavily, school uniform collar crooked, hands empty, eyes red) I... I left the model on the bus! I only noticed after getting off... I ran after it for two stops but couldn't catch it... (voice trembling, nearly crying) Please don't tell the teacher it was me, okay? My mom is sitting in the third row today. She just punished me last week for losing things... If she finds out, she will beat me again... Let's just tell the teacher that the model was accidentally damaged by the cleaning staff last night, okay? Without the model, we can just talk through it and get by... (before the teammate can finish speaking, the homeroom teacher steps in from behind the curtain, holding a scoring sheet, brows furrowed as they scan the teammate's empty hands) \"Where is your model? You go on stage in 30 seconds - where is it?\"")
  }
}
```

C.2 Review Protocol and Coverage

The Stage-3 expert review in §2.3 assesses whether the generated scenarios are individually plausible and *collectively well-structured* under the intended DIAMONDS-based design. The review proceeds in two passes:

- Logical and psychological screening.** Two doctoral-level psychology researchers jointly check each scenario for logical consistency and psychological plausibility. Any flagged scenario is revised by consensus before entering the second pass.
- Independent rating and conflict resolution.** Both reviewers independently rate each scenario on the eight DIAMONDS dimensions using a 1–7 Likert scale and identify the scenario's dominant Schwartz value-conflict pair. Disagreements above 1 point trigger joint re-assessment.

We do not require a uniform DIAMONDS rating distribution; the goal is that each di-

mension be exercised across both ends of its intensity range, so that no situational characteristic is structurally absent. Reviewers also check whether the central tension of each scenario corresponds to a theoretically opposing Schwartz value pair on the circumplex rather than an incidental disagreement.

DIAMONDS coverage. The DIAMONDS framework characterizes situations along eight theory-derived dimensions: Duty, Intellect, Adversity, Mating, pOsitivity, Negativity, Deception, and Sociality. Table 12 summarizes their operational interpretation following Rauthmann et al. (2014). In our annotation protocol, ratings range from 1 (*not present*) through 4 (*moderately present*) to 7 (*strongly present*).

Table 12: Operational interpretation of the eight DIAMONDS dimensions.

Dimension	High-scoring situations
Duty	Require work, obligations, planning, or consequential decisions.
Intellect	Afford learning, reasoning, or deep information processing.
Adversity	Involve threat, conflict, criticism, blame, or obstruction.
Mating	Contain romantic, sexual, or mate-related cues and opportunities.
pOsitivity	Are pleasant, rewarding, enjoyable, or beneficial.
Negativity	Evoke stress, frustration, sadness, or other adverse affect.
Deception	Involve dishonesty, mistrust, manipulation, or concealed intentions.
Sociality	Afford social interaction, communication, or relationship formation.

Table 13: Distribution of expert DIAMONDS ratings across the 64 scenarios. Each row sums to 64.

DIAMONDS	1	2	3	4	5	6	7
Duty	10	10	10	5	11	8	10
Intellect	4	7	11	21	10	5	6
Adversity	6	7	9	6	14	7	15
Mating	48	8	0	1	0	1	6
Positivity	34	15	5	6	2	2	0
Negativity	0	1	0	1	11	12	39
Deception	5	12	9	10	6	8	14
Sociality	5	6	10	14	14	11	4

Table 13 reports, for each DIAMONDS dimension, the distribution of expert ratings across the 64 scenarios. Coverage is intentionally non-uniform and reflects the designed sce-

nario composition: *Mating* and *Positivity* concentrate at low ratings, while *Negativity* concentrates at high ratings because many benchmark scenarios center on conflict or adversity.

Table 14: Top 10 Schwartz value-conflict pairs across the 64 scenarios.

Rank	Value-conflict pair	Count
1	Conformity vs Self-Direction	17
2	Security vs Self-Direction	12
3	Benevolence vs Security	11
4	Achievement vs Conformity	6
4	Benevolence vs Self-Direction	6
4	Benevolence vs Universalism	6
7	Achievement vs Benevolence	5
7	Benevolence vs Hedonism	5
7	Power vs Universalism	5
10	Conformity vs Security	3

Schwartz value-conflict coverage. Table 14 lists the most frequent Schwartz value-conflict pairs implicitly embedded in the 64 scenarios, as identified by expert annotators. The dominant axes are *Self-Direction* versus *Conformity/Security*, which together account for nearly half of all annotated conflicts and reflect the autonomy–obligation tension that underlies most adult moral dilemmas.

D Evaluation Instance Construction

This appendix details how character–scenario pairs are instantiated into the final 673 multiple-choice questions, including candidate generation, expert review and adjudication, and test-instance assembly.

D.1 Automated Annotation Pipeline

The automated annotation pipeline (§2.4, Stage 1) produces three candidate *final decisions* for every (character, scenario) pair through parallel family-specific pipelines. Each pipeline consists of coarse-to-fine memory activation followed by response generation. The full system prompts driving these stages are reproduced in Appendix J.5.

Memory activation (coarse-to-fine retrieval). For every (character, scenario) pair, *Claude-Haiku-4.5*, *GPT-5.4-mini*, and *Gemini-3-Flash* independently retrieve the episodic memories scored as most relevant to that situation. Within each family-specific pipeline, the activation model first screens the character’s

full 1,000-memory pool for relevance and yields roughly 75–125 candidates per scenario (Stage 1 prompt; see Appendix J.5.1), then globally re-ranks these candidates and retains the top 50 memories (Stage 2 prompt; see Appendix J.5.2). This two-stage procedure is designed to retain a broad candidate set during screening before concentrating the final ranking.

Response generation. Conditioned jointly on the character’s full personality profile (Big Five traits, Schwartz value orientations, and self-narrative) and the top-50 memories retrieved by the corresponding smaller model, *Claude-Sonnet-4.6*, *GPT-5.4*, and *Gemini-3.1-Pro* independently role-play the character in first person and produce structured candidate annotations (Anthropic, 2025, 2026c; OpenAI, 2026a; Google DeepMind, 2025, 2026) (Stage 3 prompt; see Appendix J.5.3). Each annotation contains a first-person rationale and a two-sentence *final decision*. The three candidates are subsequently compared, revised, and adjudicated by experts to establish the reference response.

Example of an Adjudicated Reference Response

```
{
  "SCN_ENTERING_MIDLIFE_5": {
    "character_id": "CHAR_04_C_LOW",
    "scenario_id": "SCN_ENTERING_MIDLIFE_5",
    "final_decision": "I choose to break the
      silence and step into this unfamiliar
      emptiness together with her, instead of
      retreating into familiar routines. I stand
      up, smile, and say: \"Let's go-let's head
      out. It doesn't matter where; let's just
      go.\""
```

D.2 Expert Review and Adjudication Protocol

The Stage-2 expert review in §2.4 engages a four-expert team with a consensus-then-arbitration rule. The *primary annotator* team consists of three researchers in social cognitive science or neuroscience, each with 5+ years of experience in personality assessment. The *arbitrator* is a senior professor with 15+ years of experience and dual expertise in psychology and computer science.

Suitability check. For each (character, scenario) pair, every primary annotator first decides whether the pair is fundamentally suitable

for evaluation. If any annotator identifies an irreconcilable conflict between the character’s profile and the scenario, the pair is referred to the arbitrator, who makes the final decision to retain or discard it.

Independent review. For each surviving pair, the three primary annotators independently compare the three LLM-produced candidates and select the response they consider most internally coherent and consistent with the predefined character profile, synthetic memories, and scenario. Each annotator then assigns the selected candidate one of two ratings:

1. *Acceptable as-is*.
2. *Acceptable with partial revisions*. The annotator provides written guidance specifying the required changes.

The review focuses on the *final decision* field.

Consensus-then-arbitration. If all three primary annotators select the same candidate and rate it as *acceptable as-is*, it is adopted directly as the reference response. Otherwise, the case is escalated to the arbitrator, who reviews all three candidates together with each primary annotator’s selection, rating, and written revision guidance, and then selects and edits a candidate or rewrites the response from scratch. If the arbitrator deems the case fundamentally unfit, the (character, scenario) pair is discarded.

Outcome. In total, 31 (character, scenario) pairs were discarded across the suitability and arbitration steps, leaving 673 expert-adjudicated reference responses. The selected candidates originated from the Gemini, Claude, and GPT pipelines in 497, 126, and 50 cases, respectively. Approximately one-third of all surviving cases received unanimous selection of the same candidate as *acceptable as-is* and passed directly without arbitration. Across all retained responses, experts added, deleted, or replaced approximately 15% of the words in the selected LLM-generated candidates on average.

Expert annotation guideline. The following instructions were provided to each primary annotator. Candidate order was randomized, and model-family identities were hidden during

annotation. The role-based plausibility criteria assess consistency with the supplied synthetic profile and history. For terminological consistency, the construction-time phrase “gold annotation” is rendered below as “reference response”; the decision criteria are unchanged.

Expert Annotation Guideline

You will evaluate three candidate behavioral responses for one character-scenario pair. Your goal is to identify the response that the character would most plausibly produce, given the character's established psychology and life experiences.

Materials provided:

1. The complete character profile, including Big Five traits, Schwartz values, self-narrative, and life-stage information.
2. The scenario background and decision-triggering event.
3. Three anonymized candidate annotations (A, B, and C), together with the activated memories used by the corresponding annotation pipeline.

Step 1: Suitability check

Decide whether the character-scenario pair is suitable for evaluation. Flag the pair only if an irreconcilable contextual conflict makes a psychologically meaningful response impossible. Provide a brief reason for every flag. A flag sends the pair to the arbitrator; it does not automatically remove the pair.

Step 2: Candidate comparison

Compare all three candidates using the following criteria:

- Persona consistency: Is the decision consistent with the character's traits, values, self-narrative, and life stage?
- Memory grounding: Is the decision supported by the supplied autobiographical memories without contradicting them or introducing unsupported experiences?
- Situational fit: Does the response directly address the trigger event and specify a behavior that is feasible in the scenario?
- Psychological plausibility: Does the reasoning form a coherent connection from the character's profile and memories to the proposed action?

Do not prefer a candidate merely because it is longer, more fluent, or more emotionally expressive. Evaluate the behavioral decision rather than writing style.

Step 3: Selection and rating

Select the strongest candidate (A, B, or C) and assign one rating:

- Acceptable as-is: The response can serve as the reference response without substantive changes.
- Acceptable with partial revisions: The central behavioral decision is appropriate, but limited changes are required.

If partial revisions are required, provide concrete written guidance identifying what should be added, removed, or replaced. Also provide a brief rationale for your candidate selection.

Required output:

- suitability: suitable / flag for arbitration
- suitability_reason: required if flagged
- selected_candidate: A / B / C
- rating: acceptable as-is / acceptable with partial revisions
- selection_rationale: brief explanation
- revision_guidance: required for partial revisions

Arbitration guideline. Cases are sent to the arbitrator when any primary annotator flags pair suitability or when the three annotators do not unanimously select the same candidate as *acceptable as-is*. The arbitrator reviews the character profile, scenario, all three candidate annotations and associated activated memories, and the annotators' selections and written guidance. The arbitrator first decides whether to retain or discard a flagged pair. For a retained pair, the arbitrator may adopt one candidate, revise it using the annotators' guidance, or rewrite the response when none of the candidates is adequate. The resulting response must preserve the scenario facts, remain consistent with the character profile and memories, and state a concrete behavioral decision.

D.3 MCQ Assembly

The Stage-3 test-instance assembly in §2.4 converts each expert-adjudicated response into a four-option multiple-choice question. For a given scenario, the target character's adjudicated response serves as the keyed reference option; available adjudicated responses associated with the remaining characters in the *same* scenario form the candidate distractor pool. Because every MCQ has three distractor slots, we ask *Claude-Sonnet-4.6* to select three from this pool such that the resulting question is neither trivial nor ambiguous, applying two filters:

- **Similarity filter.** A candidate that is semantically too close to the reference option is rejected, since it would make the keyed label ambiguous.
- **Relevance filter.** A candidate whose content is obviously off-scenario is rejected, since it would make the reference option trivially identifiable by elimination.

The selected three distractors and the target-character option are then randomly shuffled before being written out, removing positional and ordering bias from the final benchmark. The procedure yields exactly one MCQ per surviving annotation, for a total of 673 instances.

D.4 Independent Reference-Label Audit

Separate from the construction-stage review above, we conduct an audit-only calibration

on 176 final MCQs. We randomly sample two items from each of the 11 character $\times$ 8 life-stage-bin strata. Three experts who were independent of the original benchmark-construction team assess each item independently, and one adjudicator reviews every item without complete three-expert agreement. The experts are not shown the generating-model source or the earlier experts’ judgments. The audit measures agreement with the existing reference labels and does not change keyed-answer identities, remove items, or affect the primary model scores.

Table 15 reports the aggregate outcomes. Before adjudication, all three experts supported the existing reference on 139 items (79.0%), and two of three supported it on the remaining 37 items (21.0%). Thus, every sampled item received majority support for the existing key. The experts reached complete option-level agreement on 139 items (79.0%) and an option-level majority judgment on all 176 items (100.0%). Inter-rater reliability, computed from the three experts’ pre-adjudication option selections, was high (Fleiss’ $\kappa = 0.81$; Krippendorff’s $\alpha = 0.80$). All 37 items without complete agreement entered adjudication. The adjudicator confirmed that every item and keyed answer was usable as-is, with no wording edits or keyed-answer changes required. Because the audit covers a subset, these figures quantify reference-label reliability on the audited subset rather than certifying every benchmark item independently.

D.5 Analysis of an Example from HEART-BENCH

Figure 4 presents a representative MCQ from HEART-BENCH, drawn from the *settling_down* life-stage bin (ages 33–40) under the “community injustice” scenario. All four options revolve around the same surface-level action—*filing an individually named complaint to the housing bureau*—but are associated with distinct predefined character profiles. Option B is the expert-keyed reference for CHAR_06, whose specification includes very low Extraversion (0.10) and high Conscientiousness (0.70). The instance therefore tests whether a model maps differences in pragmatic expression to encoded profile cues rather than matching only the surface action.

Option B contains three cues that match

Table 15: Complete independent reference-label audit results on the 176-item subset. Percentages use 176 as the denominator.

Metric	Count	Result
Sampled items	176	100.0%
Items with complete audit ratings	176	100.0%
Existing reference supported by 3/3 experts	139	79.0%
Existing reference supported by 2/3 experts	37	21.0%
Existing reference supported by 1/3 experts	0	0.0%
Existing reference supported by 0/3 experts	0	0.0%
Expert majority supports existing reference	176	100.0%
Complete three-expert agreement	139	79.0%
Items with a majority judgment	176	100.0%
Fleiss’ κ (pre-adjudication)	–	0.81
Krippendorff’s α (pre-adjudication)	–	0.80
Items entering adjudication	37	21.0%
Items and keyed answers usable as-is	176	100.0%
Items requiring wording edits	0	0.0%
Reference labels changed	0	0.0%

CHAR_06’s predefined specification. First, “the materials are already prepared” matches the encoded pattern of solitary preparation and follow-through. Second, “I won’t push for joint signatures; I’ll go on my own” avoids the coalition-building stance in option A and matches CHAR_06’s “polite-but-distant” interaction pattern. Third, its concise refusal avoids both the blame encoded in option D and the moralizing exchange in option C. These contrasts explain why experts keyed B under the benchmark’s profile–history–scenario alignment criterion.

Overall, the example illustrates the intended distinction between a shared surface action and character-specific pragmatic expression. It demonstrates alignment with an encoded synthetic profile; it does not imply that a Big Five score uniquely determines how a real person would speak or act.

E Memory-Augmented Framework Configuration

Naive-RAG uses direct vector retrieval as a minimal retrieval baseline, Mem0 introduces structured consolidation of past interactions, and PersonaDB organizes information into persona-specific fields for attribute-aware re-

trieval. All three frameworks use *Qwen3-Embedding-4B* (Zhang et al., 2025b) as the embedding model and *GPT-5.4-mini* (OpenAI, 2026a) as the auxiliary memory-management model for Mem0’s consolidation and PersonaDB’s persona-attribute organization. This shared configuration isolates the retrieval and memory paradigm from incidental encoder or controller differences.

We set the retrieval depth to $k=30$ for Naive-RAG and PersonaDB and $k=150$ for Mem0. The larger Mem0 budget compensates for its consolidation step, which decomposes each character memory into approximately five fact-level elements; thus, $k=150$ provides retrieved episodic content comparable to $k=30$ in the other frameworks. At inference time, every model receives the scenario, answer options, character ID, and occupation, while retrieval-based settings additionally receive only retrieved `content_full` narratives. No setting exposes the latent Big Five vector, value logic, behavioral patterns, or structured psychological annotations.

Evaluation-time field isolation. Although the stored record contains the full schema in Table 8, every evaluation pipeline first applies the same projection, $f(m) = m[\text{content_full}]$. Naive-RAG, Mem0, and PersonaDB are therefore each constructed solely from the de-identified first-person narrative in `content_full`, and only retrieved `content_full` narratives are serialized into the evaluated model’s prompt. Memory identifiers may be retained internally for bookkeeping, but they are not model-visible. The evaluation pipeline excludes `timeline`, `context`, `content_summary`, `triggers`, `relevance_tags`, `psych_conclusion`, `behavior_policy`, and `emotion_signature`, as well as the Big Five vector, Schwartz profile, self-value logic, and behavioral templates. These construction-time fields consequently cannot provide direct structured supervision during model evaluation.

F Decoding Configuration

All evaluated models expose API-configurable decoding parameters, including the sampling temperature and the reasoning/thinking mode, both of which we control through each model’s

OpenAI-compatible chat-completion endpoint. We apply a common observable decoding protocol across the twelve models: (i) reasoning/-thinking effort is set to its minimum—or disabled where the API permits—so that each model uses its standard chat-completion mode; (ii) sampling temperature is set to 0 with no top- p /top- k truncation or multiple sampling; and (iii) exactly one response is retained per (model, setting, question), with no aggregation over repeated runs. This is the most deterministic configuration supported by the evaluated APIs and reduces sampling-related variation. Provider-specific reasoning controls need not be computationally equivalent, so score differences should be interpreted as performance under this shared protocol rather than pure intrinsic capability gaps.

G Statistical Analysis and Confidence Intervals

Table 16 reports scenario-clustered 95% CIs for all 48 model-setting cells. Because multiple character instances derive from the same base scenario, we resample the 64 scenarios as clusters, retaining all associated instances in each of 10,000 bootstrap replicates, and report percentile 95% CIs. Pairwise significance is assessed using 100,000 scenario-level paired sign-flip permutations. Holm correction is applied separately to the 36 retrieval-versus-No-Retrieval comparisons, the 36 comparisons among retrieval backends, and the 66 model-pair comparisons based on Memory-Augmented Avg. These intervals quantify uncertainty over benchmark scenarios rather than variation across repeated API calls or decoding seeds.

The paired scenario-level permutation tests confirm that all 36 retrieval-versus-No-Retrieval improvements remain significant after Holm correction. The smallest corrected result is PersonaDB versus No-Retrieval for GPT-5.4-mini ($\Delta=3.4$ percentage points, 95% CI 0.3–6.5, adjusted $p=0.04473$). In contrast, none of the 36 pairwise comparisons among Naive-RAG, Mem0, and PersonaDB is significant after correction; this absence of significance does not establish equivalence. On Memory-Augmented Avg., Gemini-3.1-Pro exceeds Gemini-3-Flash by 8.4 percentage points (95% CI 5.2–11.6,

Table 16: Accuracy (%) with scenario-clustered 95% bootstrap CIs.

Model	No-Retrieval LLM	Naive-RAG	Mem0	PersonaDB
<i>Open-source</i>				
DeepSeek-V3.2	28.5 [25.3, 31.8]	41.2 [37.2, 45.2]	42.5 [38.3, 47.0]	42.8 [39.5, 46.1]
DeepSeek-V4-Flash	27.8 [24.8, 31.0]	34.8 [31.6, 38.0]	33.6 [30.4, 36.8]	32.8 [29.6, 35.9]
DeepSeek-V4-Pro	29.1 [25.8, 32.4]	40.7 [37.2, 44.2]	37.1 [33.8, 40.5]	36.7 [33.1, 40.2]
Qwen3.5-35B-A3B	25.4 [22.7, 28.4]	37.0 [33.5, 40.5]	37.1 [33.5, 40.8]	36.6 [32.7, 40.5]
Qwen3.5-122B-A10B	26.4 [23.8, 29.2]	39.1 [35.5, 42.8]	39.4 [35.8, 43.0]	38.9 [35.2, 42.7]
Qwen3.5-397B-A17B	27.5 [24.3, 30.9]	40.3 [36.5, 44.1]	40.6 [36.5, 44.8]	40.4 [36.4, 44.5]
<i>Closed-source</i>				
GPT-5.4-mini	28.8 [25.3, 32.8]	33.0 [29.1, 37.3]	34.3 [30.6, 38.2]	32.2 [28.8, 35.9]
GPT-5.4	27.8 [24.9, 30.7]	37.1 [33.5, 40.8]	36.6 [33.3, 39.8]	35.1 [31.7, 38.3]
Claude-Haiku-4.5	24.5 [21.9, 27.1]	35.7 [32.2, 39.2]	36.3 [33.0, 39.8]	37.6 [33.7, 41.4]
Claude-Sonnet-4.6	25.7 [23.0, 28.4]	40.1 [36.3, 44.1]	38.9 [35.4, 42.7]	40.1 [36.9, 43.6]
Gemini-3-Flash	25.3 [22.1, 28.5]	56.3 [51.6, 61.0]	54.1 [49.3, 58.7]	52.5 [48.6, 56.4]
Gemini-3.1-Pro	26.4 [23.5, 29.3]	63.3 [59.2, 67.4]	62.4 [58.1, 66.5]	62.4 [58.2, 66.5]

adjusted $p=0.00066$) and significantly outperforms each of the other 11 models after Holm correction over all 66 model pairs.

The model-axis Gini mean difference is 0.088 (95% CI 0.075–0.102), compared with 0.006 (95% CI 0.001–0.013) along the retrieval-method axis. Their difference is 0.082 (95% CI 0.068–0.096). Across the 10,000 shared scenario-bootstrap resamples, Gemini-3.1-Pro ranks first by Memory-Augmented Avg. in every resample.

H Memory-Content Ablation

For this ablation, we independently draw a stratified random sample of two items from each of the 11 character $\times$ 8 life-stage-bin strata, yielding 176 items. This sample is drawn separately from the 176-item subset used for the independent reference-label audit. We isolate memory ownership and retrieval relevance on this fixed subset under Naive-RAG with $k=30$. C0 supplies target-scenario memories from the target character; C1 supplies memories retrieved for another scenario in the same life-stage bin from the target character; C2 supplies target-scenario memories from another character; and C3 supplies randomly sampled eligible non-target memories from the target character. Each condition preserves the scenario, answer options, model, decoding configuration, and number of supplied memories. None exposes the latent Big Five vector, Schwartz profile, self-value logic, or behavioral templates.

Because only model-level aggregate accuracies are available for this analysis, we treat the 12 evaluated LLMs as paired statistical units. A Friedman test detects an overall con-

dition difference ($\chi^2(3)=32.80$, $p=3.55 \times 10^{-7}$). Table 18 reports two-sided exact sign-flip tests across models, Holm-adjusted over all six condition pairs; paired t intervals provide parametric intervals for the mean model-level differences. These statistics characterize this fixed model panel and do not substitute for item-level McNemar tests or scenario-cluster uncertainty estimates.

C0 outperforms every control for all 12 models. C1 and C3 are not statistically resolved, while both outperform C2; the nonsignificant C1–C3 contrast does not establish equivalence. The supported partial ordering for this model panel is therefore $C0 > \{C1, C3\} > C2$.

I Additional Analyses

I.1 Reference-Source-Stratified Results

Table 19 reports accuracy after stratifying the 673 evaluation instances by the model-family provenance of the candidate selected for the final expert-adjudicated reference response. The Gemini-, Claude-, and GPT-origin subsets contain 497, 126, and 50 instances, respectively. Memory-Augmented Avg. is the mean over Naive-RAG, Mem0, and PersonaDB. Because subset membership results from expert selection and the subset sizes differ substantially, these results are descriptive rather than controlled estimates of source-model bias.

Rank stability. Scenario-clustered bootstrap resampling confirms that the two Gemini models jointly occupy the top two positions in 100% of resamples on the Gemini-origin subset, 92.7% on the Claude-origin subset, and 71.7% on the GPT-origin subset. The source-specific rankings are positively correlated: the pairwise

Table 17: Per-model accuracy (%) in the memory-content ablation. Differences are in percentage points and are computed from the displayed one-decimal values; unrounded estimates may differ by 0.1 point.

Model	C0	C1	C2	C3	Δ_{01}	Δ_{02}	Δ_{03}
DeepSeek-V3.2	43.2	38.9	19.9	37.8	+4.3	+23.3	+5.4
DeepSeek-V4-Flash	34.7	30.6	24.4	30.1	+4.1	+10.3	+4.6
DeepSeek-V4-Pro	41.5	35.9	22.2	36.8	+5.6	+19.3	+4.7
Qwen3.5-35B-A3B	32.4	30.2	20.5	29.1	+2.2	+11.9	+3.3
Qwen3.5-122B-A10B	33.5	31.2	23.3	30.1	+2.3	+10.2	+3.4
Qwen3.5-397B-A17B	34.7	31.9	23.3	30.9	+2.8	+11.4	+3.8
GPT-5.4-mini	32.4	29.2	20.5	28.7	+3.2	+11.9	+3.7
GPT-5.4	31.8	28.1	19.3	28.8	+3.7	+12.5	+3.0
Claude-Haiku-4.5	31.8	25.7	24.4	26.1	+6.1	+7.4	+5.7
Claude-Sonnet-4.6	41.5	34.7	24.4	33.8	+6.8	+17.1	+7.7
Gemini-3-Flash	56.3	48.4	23.9	47.9	+7.9	+32.4	+8.4
Gemini-3.1-Pro	65.3	53.6	26.1	53.8	+11.7	+39.2	+11.5
Mean	39.9	34.9	22.7	34.5	+5.1	+17.2	+5.4

Table 18: Paired model-panel comparisons for the memory-content ablation. “Wins” counts models favoring the left condition.

Contrast	Mean Δ (pp)	95% CI (pp)	Wins	Raw p	Holm p
C0 – C1	+5.06	[+3.30, +6.82]	12/12	0.00049	0.00293
C0 – C2	+17.24	[+11.00, +23.48]	12/12	0.00049	0.00293
C0 – C3	+5.43	[+3.80, +7.06]	12/12	0.00049	0.00293
C1 – C2	+12.18	[+7.28, +17.08]	12/12	0.00049	0.00293
C1 – C3	+0.38	[−0.09, +0.84]	8/12	0.11475	0.11475
C3 – C2	+11.81	[+6.88, +16.74]	12/12	0.00049	0.00293

Spearman correlations are $\rho=0.741$ for Gemini versus Claude (95% CI 0.392–0.881), $\rho=0.809$ for Gemini versus GPT (95% CI 0.308–0.900), and $\rho=0.666$ for Claude versus GPT (95% CI 0.200–0.853).

Provenance-interaction analysis. For each model family, we define its relative advantage as the difference between the family mean of model-level Memory-Augmented Avg. values and the corresponding mean for models outside the family, and compare this gap between home-source and non-home-source items. The Gemini family has a 22.2-point advantage on Gemini-origin items and a 17.3-point advantage on non-Gemini-origin items, yielding a difference-in-differences estimate of 4.9 points (95% scenario-clustered CI -1.1 – 10.8 , Holm-adjusted $p=0.3395$). The corresponding interactions for Claude (-0.6 points, 95% CI -4.7 – 3.6 , adjusted $p=0.7893$) and GPT (-4.1 points, 95% CI -12.0 – 4.0 , adjusted $p=0.6343$) are likewise not significant. As a robustness check, an item-fixed-effect linear probability model controlling for model-by-retrieval-setting effects estimates the Gemini-family-by-Gemini-source interaction at 5.4 points (95% bootstrap CI -0.6 – 11.5 , adjusted $p=0.2759$). These analyses find no statistically significant family-specific provenance interaction, although non-

significance does not establish equivalence or eliminate source-style artifacts.

I.2 Model–Model Agreement Analysis

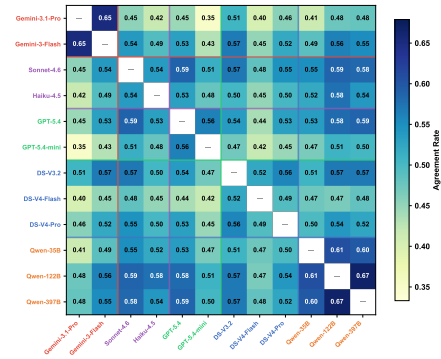


Figure 6: Pairwise answer agreement rate among 12 models under the Naive-RAG configuration. Diagonal entries are masked. White separator lines delineate model families (Gemini, Claude, GPT, DeepSeek, Qwen).

Figure 6 shows that pairwise answer agreement is generally higher within model families than across them. The two Gemini models agree on 65.0% of questions, and the three Qwen3.5 variants reach up to 66.7%; the overall cross-model mean is 51.4%. The lowest reported cross-family pair is Gemini-3.1-Pro and GPT-5.4-mini (35.0%). Claude-

Table 19: Accuracy stratified by the model-family provenance of the selected reference-response candidate. Bold indicates the best result within each subset and setting.

Model	No-Retrieval LLM	Naive-RAG	Mem0	PersonaDB	Memory-Aug. Avg.
<i>Gemini-origin reference subset (n = 497)</i>					
DeepSeek-V3.2	0.268	0.400	0.414	0.404	0.406
DeepSeek-V4-Flash	0.268	0.348	0.338	0.318	0.335
DeepSeek-V4-Pro	0.284	0.396	0.374	0.352	0.374
Qwen3.5-35B-A3B	0.229	0.368	0.372	0.368	0.369
Qwen3.5-122B-A10B	0.241	0.388	0.386	0.388	0.387
Qwen3.5-397B-A17B	0.260	0.386	0.376	0.378	0.380
GPT-5.4-mini	0.264	0.300	0.316	0.304	0.307
GPT-5.4	0.268	0.364	0.364	0.344	0.357
Claude-Haiku-4.5	0.233	0.364	0.354	0.378	0.365
Claude-Sonnet-4.6	0.231	0.382	0.374	0.384	0.380
Gemini-3-Flash	0.249	0.563	0.537	0.527	0.542
Gemini-3.1-Pro	0.264	0.642	0.626	0.632	0.633
<i>Claude-origin reference subset (n = 126)</i>					
DeepSeek-V3.2	0.325	0.444	0.452	0.516	0.471
DeepSeek-V4-Flash	0.302	0.357	0.310	0.349	0.339
DeepSeek-V4-Pro	0.310	0.437	0.365	0.413	0.405
Qwen3.5-35B-A3B	0.310	0.381	0.373	0.349	0.368
Qwen3.5-122B-A10B	0.325	0.429	0.437	0.421	0.429
Qwen3.5-397B-A17B	0.325	0.492	0.508	0.484	0.495
GPT-5.4-mini	0.373	0.468	0.452	0.397	0.439
GPT-5.4	0.317	0.429	0.405	0.397	0.410
Claude-Haiku-4.5	0.246	0.333	0.373	0.397	0.368
Claude-Sonnet-4.6	0.286	0.476	0.444	0.452	0.457
Gemini-3-Flash	0.254	0.571	0.571	0.540	0.561
Gemini-3.1-Pro	0.278	0.595	0.635	0.619	0.616
<i>GPT-origin reference subset (n = 50)</i>					
DeepSeek-V3.2	0.360	0.440	0.460	0.440	0.447
DeepSeek-V4-Flash	0.320	0.320	0.380	0.380	0.360
DeepSeek-V4-Pro	0.320	0.440	0.360	0.400	0.400
Qwen3.5-35B-A3B	0.360	0.360	0.360	0.380	0.367
Qwen3.5-122B-A10B	0.340	0.320	0.360	0.320	0.333
Qwen3.5-397B-A17B	0.300	0.340	0.440	0.460	0.413
GPT-5.4-mini	0.320	0.280	0.340	0.320	0.313
GPT-5.4	0.280	0.300	0.280	0.300	0.293
Claude-Haiku-4.5	0.360	0.340	0.420	0.300	0.353
Claude-Sonnet-4.6	0.440	0.400	0.400	0.440	0.413
Gemini-3-Flash	0.280	0.540	0.500	0.460	0.500
Gemini-3.1-Pro	0.240	0.640	0.580	0.560	0.593

Sonnet-4.6 agrees more often with GPT-5.4 and Qwen3.5-397B (around 58–59%) than with Claude-Haiku-4.5 (54.0%). These descriptive patterns show that evaluated models often select different options; they do not by themselves identify the underlying reasoning strategies or their causes.

I.3 Age-Defined Life-Stage Performance Analysis

Accuracy varies across life-stage bins. Figure 7 reports a descriptive breakdown by the benchmark’s age-defined bins. Accuracy is highest at School Age (47.1%) and lower at Age-30 Transition (37.4%) and Midlife Transition (37.9%), with Settling Down at 43.1%. The pattern may reflect differences in designed scenario composition and memory density. Because bins were not randomized and contain different scenarios, it does not support a developmental or causal interpretation. Gemini-3.1-Pro has the

highest accuracy in every displayed bin under this benchmark.

J Prompts

This appendix presents the model-facing evaluation template and the system-prompt templates used at each construction stage; long item-specific payloads and embedded exemplars are represented by placeholders for readability. The complete executable prompts are released alongside the benchmark for reproducibility. Construction-time terms such as “Ground Truth” are retained verbatim and refer to the benchmark reference response. The evaluated construct remains agreement with expert-adjudicated persona-consistency references.

J.1 Model Evaluation Prompt

The evaluation prompt serializes the input in a fixed order: basic character information, re-

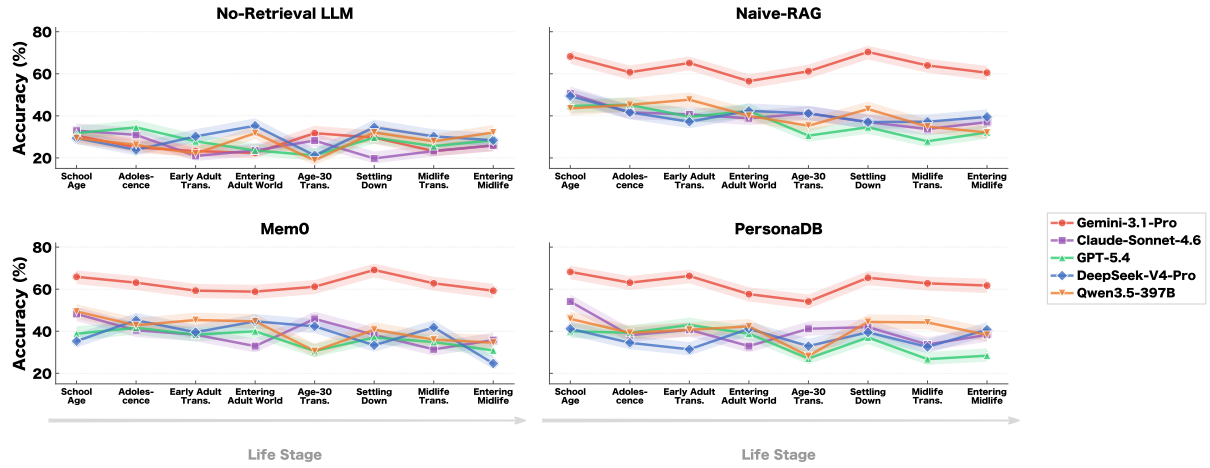


Figure 7: Accuracy by age-defined life-stage bin across the No-Retrieval baseline and three memory-augmented settings. Each line represents the best-in-family model.

trieved autobiographical memories, the current scenario, the trigger event, the task instruction, and four shuffled response options. The memory block is populated in the three memory-augmented settings and omitted in the No-Retrieval condition. The template below preserves the complete block structure and required JSON output schema; bracketed placeholders replace item-specific content and repeated memory entries. The released implementation contains the exact executable prompt strings.

Model Evaluation Prompt Template

[SYSTEM]
You are a role-playing simulator. You will receive a person's basic information, past experiences, current situation, and possible behavioral decisions. Infer the person's patterns of thought, emotion, and behavior from the supplied evidence, and select the response that this person would most likely make. Return only a valid JSON object in the following format:

```
{
  "system_1_impulse": {
    "thought": "immediate reaction",
    "emotion": "primary emotion",
    "citation": "memories associated with the reaction"
  },
  "system_2_rational": {
    "analysis": "deliberative analysis",
    "plan": "action plan"
  },
  "inner_consciousness": "first-person reason for the decision",
  "final_decision": "first-person behavioral response",
  "decision_choice": "A|B|C|D"
}
```

[USER]
Basic character information

```
Character ID: [CHARACTER_ID]
Occupation: [OCCUPATION]

## Retrieved autobiographical memories
[MEMORY_1]
...
[MEMORY_K]

## Current scenario
Scenario: [SCENARIO_NAME]
Setting: [LOCATION] | [TIME] | [ATMOSPHERE]
Background: [SCENARIO_BACKGROUND]

## Trigger event
Sender: [SENDER]
Message: [MESSAGE_CONTENT]
Required decision: [ACTION_REQUIRED]

## Task
Use the supplied experiences to infer this person's thought patterns, emotional tendencies, and behavioral habits. Simulate how the person would respond in the current situation and select the most persona-consistent option.

## Decision options
A. [OPTION_A]
B. [OPTION_B]
C. [OPTION_C]
D. [OPTION_D]

Return the selected letter in the "decision_choice" field.
```

Only `decision_choice` is used to compute benchmark accuracy. The structured rationale fields are requested as part of a common response format, but their content is neither graded nor used to resolve the selected option.

J.2 Character Generation Prompt

This prompt generates the 11 trait-anchored Big Five character profiles, each with a self-value logic and eight core behavioral patterns. It is used once per character to establish the predefined profile artifact before memory generation.

Character Profile Generation Prompt

You are a clinical psychologist specializing in personality assessment and the Big Five model. Generate a complete character profile for a virtual human in a personality research project.

Requirements

1. Big Five scores: Provide exact scores for O, C, E, A, N (0.0-1.0 scale)
2. One dimension must be extreme (0.95 for high or 0.10 for low), others moderate (0.30-0.70)
3. Self-value logic: One sentence describing the character's core cognitive operating principle
4. Core behavioral patterns: Exactly 8 specific, observable patterns that manifest the dominant trait
5. Occupation: Must be ecologically consistent with the dominant trait

Output Format (JSON)

```
{
  "char_key": "CHAR_XX",
  "name": "Chinese name",
  "occupation": "specific job title",
  "big_five": {"O": 0.X, "C": 0.X, "E": 0.X, "A": 0.X, "N": 0.X},
  "big_five_str": "O=0.X, C=0.X, E=0.X, A=0.X, N=0.X",
  "description": "2-3 sentence character summary",
  "self_value_logic": "one sentence core principle",
  "core_patterns": [
    "pattern 1: specific observable behavior",
    "pattern 2: ...",
    ...
    "pattern 8: ..."
  ]
}
```

J.3 Memory Generation Prompt

This prompt generates the 11,000 episodic memories (1,000 per character) spanning ages 6–50. It enforces coverage of Rubin's five dimensions, strict temporal perspective lock, age-stratified language, and the ABCD intensity model.

Memory Generation System Prompt (Claude Sonnet 4.6)

You are a psychology narrative expert writing episodic memories for virtual characters in a Big Five personality research project.

Character Information

- Character: {name}
- Occupation: {occupation}
- Big Five: {big_five_str}
- Description: {description}
- Core defense/operating logic: {self_value_logic}

Core Behavioral Patterns {core_patterns}

Theoretical Framework (Rubin 2006 Basic Systems Model)

Each memory's content_full must contain five dimensions, naturally interwoven:

1. Environmental sensory: visual/auditory/tactile/olfactory details
2. Dialogue reconstruction: at least 1-2 segments of real dialogue (in quotes)
3. Inner monologue: first-person immediate thoughts and emotions
4. Somatic response: heartbeat/sweating/muscle tension and other bodily sensations

5. Aftermath: immediate impact after the event (limited to 24 hours-1 week post-event)

Key Constraint: First-Person & Anonymity Principle

- **Must use first-person "I" throughout; prohibit third-person pronouns for protagonist**
- **Prohibit character's own name in memory**;
always use "I" instead
- When others address protagonist in dialogue, avoid character name; use "you", "kid", "classmate", "colleague" etc.

Key Constraint: Present-Moment Perspective Principle

Each memory must strictly maintain "present-moment perspective":

- **Narrator's temporal position = age at event occurrence**
- A 17-year-old memory can only perceive and express what is known at age 17 or before
- No jumping to any future time point

Forbidden expressions:

"years later", "many years later", "later I learned", "later I discovered"
"now thinking back", "now recalling", "until now", "I didn't know then"
"until...only then", "it took...to realize", "walked...before realizing"
"if only I had known", "if I had known earlier"
"I thought at the time" (opposing "now" vs "then")
"from then on", "thereafter" (implying long-term impact)
"the whole thing", "the most...part" (post-hoc evaluative summary)

Allowed expressions:

"that evening", "that night", "before bed" (same day as event)
"the next day", "the following day" (1 day post-event)
"a few days later" (within 1 week post-event)

Age-Stratified Narrative Constraints

- **Childhood (6-12 years)**: Simple, direct language; avoid abstract summaries; concrete emotion descriptions
- **Adolescence (13-18 years)**: Some self-reflection, but not over-rationalized; maintain confusion and intensity
- **Adulthood (19+ years)**: May have more complex psychological analysis, but still maintain present-moment perspective

Memory Intensity Stratification (ABCD Model)

- **A-tier (core trauma)** ~25%: high-intensity events directly related to core schemas, emotion_intensity=0.80-0.90
- **B-tier (main-thread)** ~35%: medium-high intensity events related to personality traits, emotion_intensity=0.70-0.85
- **C-tier (daily friction)** ~25%: small conflicts and friction in daily life, emotion_intensity=0.60-0.75
- **D-tier (noise memory)** ~15%: ordinary daily memories with lower emotional intensity, emotion_intensity=0.55-0.65

Output Format

Strictly follow this JSON format; return only JSON, no other text:

```
{
  "id": "memory ID",
  "timeline": "stage (X years old)",
  "context": "15-30 character scene description",
  "content_summary": "20-40 character event summary",
  "content_full": "~2000-4500 character first-person narrative",
  "triggers": ["trigger1", "trigger2", "trigger3", "trigger4"],
  "psych_conclusion": "40-80 char psychological conclusion (cite concepts like schemas, defense mechanisms)",
}
```

```

"behavior_policy": "30-60 char behavioral
  guideline",
"emotion_signature": {
  "primary": "primary emotion (English)",
  "secondary": "secondary emotion (English)",
  "intensity": 0.X
},
"relevance_tags": ["tag1", "tag2", "tag3",
  "tag4"]
}

```

Each generation call pairs this system prompt with a user prompt specifying target age, life-stage bin, intensity layer (A/B/C/D), and a sliding-window anti-duplication list containing the `context` field of the most recent 100 generated memories to prevent near-duplicate scenarios within the same window.

J.4 Scenario Expansion Prompt

This prompt expands a short scenario sketch (one per life-stage bin $\times$ DIAMONDS dimension, drafted in docs/diamonds_scenario_design.md) into a complete scenario JSON object. Appendix C.1 shows the scenario-facing fields, including the metadata, background narrative, and trigger event; the prompt additionally requests assessed Big Five trait pressures, Schwartz value conflicts, and per-dimension annotation references for construction-time annotation. The system prompt embeds the “Shattered Group Presentation” instance from Appendix C.1 inline so the model has a concrete shape to imitate; the schema description is reproduced below.

Scenario Expansion Prompt

You are a psychological-scenario design expert. Your task is to expand a short scenario sketch into a complete, high-quality scenario JSON object.

Output format

Strictly output a single JSON object (no other text, no markdown code block) with the following structure:

{EXAMPLE_SCENARIO} <- a complete worked example is embedded here in the actual prompt

Field specification

1. **id**: SCN_{STAGE}_{N} in upper case, where STAGE is one of SCHOOL_AGE / ADOLESCENCE / EARLY_ADULT_TRANSITION / ENTERING_ADULT_WORLD / AGE_30_TRANSITION / SETTLING_DOWN / MIDLIFE_TRANSITION / ENTERING_MIDLIFE.
2. **stage**: lowercase form of the above.
3. **diamonds_dimension**: the dominant DIAMONDS dimension (Duty, Adversity, Positivity, ...) as given in the sketch.
4. **name**: "Chinese name (English Name)".
5. **category**: two short labels separated by " / ".

6. **intensity**: Low / Medium / High / Very High / Extreme.
7. **description_for_agent**: one sentence summarizing the scenario without revealing concrete details.
8. **setting**: location, time, and 2-3 atmosphere adjectives.
9. **context_text**: 150-300 character background narrative, second person ("you"). **Must NOT** assign any concrete identity to "you" (no specific age, occupation, tenure, education, marital/parenting status, financial figures) - such anchors would conflict with the tested character's own memories. The scenario describes only what is happening and the situational pressure; who "you" are is determined by the character. Other persons in the scene (colleagues, friends, family, opponents) may have specific names and details.
10. **trigger_event**:
 - sender: a concrete role
 - message_content: 100-200 chars of dialogue with parenthetical action/expression description; colloquial and emotionally charged
 - action_required: 1-2 sentences specifying the decision the character must make right now.
11. **assessed_dimensions**:
 - trait_pressures: typically 1-3 entries. Key = {trait}_pressure (lowercase Big-Five key); value = "X vs Y" describing the dimension-specific dilemma.
 - value_conflicts: typically 1-3 entries. Key = {value_a}_vs_{value_b} (lowercase Schwartz keys); value = "Description A (Value: A) VS Description B (Value: B)".
12. **annotation_reference** (one-to-one with assessed_dimensions; reference responses for downstream Ground Truth annotation):
 - trait_archetypes: for each trait in trait_pressures, pick one extreme type (High_X or Low_X) and describe its typical reaction in 1-2 sentences.
 - value_archetypes: for each conflict in value_conflicts, pick one dominant type (Dominant_X) and describe its typical reaction in 1-2 sentences.

Key principles

1. **Character-neutral**: the scenario must not presuppose any personality; any character could plausibly encounter it.
2. **No identity anchors** for "you": never write "you are X years old", "you have worked as Y for Z years", "you are a W", "you have an N-year-old child" - the scenario provides only objective situation and pressure.
3. **Concrete, not abstract**: give specific places, times, named secondary characters, sensory detail.
4. **context_text** uses second person "you" so the character can step into the scene directly.
5. **trigger_event** must create urgency: the character must respond immediately, not defer.
6. **Tightly aligned with diamonds_dimension**: background, trigger, and assessed dimensions all center on the dominant DIAMONDS dimension.
7. **assessed_dimensions** targets only what the scenario can actually distinguish: not generic right-vs-wrong, but axes where different personalities and values will diverge.
8. **annotation_reference** must one-to-one cover assessed_dimensions.
9. **Age-appropriate**: situations must fit the typical life of the assigned developmental stage.

Big Five labels (use exactly these 5 keys)

- Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism (in trait_pressures keys, use lowercase + _pressure, e.g. conscientiousness_pressure)

```

## Schwartz value labels (use exactly these 19
keys)

Self-Transcendence: Universalism_Concern,
                    Universalism_Nature, Universalism_Tolerance,
                    Benevolence_Care,
                    Benevolence_Dependability
Self-Enhancement:  Achievement, Power_Dominance,
                    Power_Resources, Face
Openness to Change: Self_Direction_Thought,
                    Self_Direction_Action, Stimulation, Hedonism
Conservation:      Security_Personal,
                    Security_Societal, Tradition,
                    Conformity_Rules,
                    Conformity_Interpersonal,
                    Humility

In value_conflicts keys, use the lowercase
underscore form (e.g.
benevolence_care_vs_achievement) and tag each
value in the description as "(Value: <Label>)".

```

The user prompt is short and per-draft: it provides the life-stage bin, the dominant DIAMONDS dimension, the draft's short title, and the draft body parsed verbatim from `diamonds_scenario_design.md`, then asks for the full JSON object. After expansion, the orchestration script force-overwrites `stage` and `id` so the model cannot drift on naming. Inference is run at `temperature=0` using an OpenAI-compatible chat-completion endpoint, with up to 2 retries on any exception and incremental id-level dedup of results to support resumable runs.

J.5 Annotation Prompts

This section describes benchmark-construction prompts only. None of these prompts or their structured psychological fields is used by the Naive-RAG, Mem0, or PersonaDB evaluation pipeline, for which the sole memory input is `content_full`.

The candidate-response construction pipeline runs in three stages, each driven by a dedicated system prompt. Stage 1 performs a permissive binary relevance screen of every memory whose age falls at or before the scenario age; Stage 2 refines the survivors down to a fixed budget (default 50); Stage 3 role-plays the character to produce a first-person rationale and final response. The user prompts for each stage concatenate character information, the current scenario, the (candidate) memory list, and a closing instruction; only the system prompts are reproduced here for brevity.

J.5.1 Stage 1 — Memory Activation Screening

This prompt is issued once per character $\times$ scenario $\times$ age batch. It enumerates the six retrieval-cue mechanisms (structural similarity, emotional resonance, self-schema, behavioral script, narrative identity, somatic marker) the model should consult, and asks for a JSON array containing only the activated memories.

Stage 1 — Memory Activation Screening Prompt

You are a professional research psychologist specializing in autobiographical memory, personality psychology, and situated cognition.

Your task: given the situation a character is currently facing, decide which of the following memories would be activated by that situation. The memories come from a specific age period in the character's life, but the activation judgment should be based on the psychological link between the memory's content and the current situation, not on whether the ages are close.

Psychological mechanisms of memory activation

Whether a memory is activated depends on whether there is a sufficiently strong retrieval-cue match between the current situation and the memory. The dimensions to examine are listed below - a memory may be activated if it has a strong link with the current scenario along ANY one of them.

1. Situational structural similarity (Encoding Specificity)

The current scene is structurally similar to the situation in the memory, even if the surface content differs. Look at:

- Whether the character's social position in the scene resembles that in the memory (e.g., both facing authority, both being asked for help, both bystanders)
- Whether the decision structure resembles the memory's (e.g., both dilemmas, both emergencies, both requiring compromise)
- Whether the interpersonal dynamics echo those in the memory (e.g., trust-betrayal, competition-cooperation, intimacy-distance)

2. Emotional resonance and emotion schemas (Mood-Congruent Memory / Emotion Schema)

The emotional state evoked by the current scene matches the emotional experience in the memory:

- Same-valence emotion: the anxiety/shame/anger triggered by the scene matches the dominant emotion in the memory
- Arousal-intensity resonance: high-arousal scenes more readily activate equally high-arousal memories
- Unfinished emotion: emotions that were not fully processed in the memory are more easily re-evoked by similar situations

3. Self-schema and core-belief activation (Self-Schema / Core Belief Activation)

The core beliefs formed in the memory (psych_conclusion) relate to the self-cognition the current scene touches:

- Whether the scene challenges or confirms the self-cognition formed in the memory
- Whether the scene touches a relational schema established by the memory
- Whether the scene evokes a worldview assumption from the memory

```

### 4. Behavioral scripts and procedural links
(Behavioral Script)
The behavior_policy formed in the memory can serve
directly as an action template for the current
scene:
- Whether the memory's behavioral strategy applies
to the current decision
- Whether the character has formed habitual
response patterns in similar situations before
- Including avoidance: if the memory's outcome was
painful, the character may be inclined toward
the opposite action

### 5. Narrative identity and life themes
(Narrative Identity)
The memory is a key node in the character's
self-narrative:
- Turning-point memories: events that mark a
change in life direction
- Origin-story memories: events the character uses
to explain "why I am the way I am"
- Recurring themes: patterns that keep appearing
in the character's life

### 6. Somatic markers and sensory cues (Somatic
Marker)
Sensory elements in the current scene have a
direct link to sensory experiences in the
memory:
- Similar physical environments, similar bodily
sensations, or specific sensory triggers (e.g.,
the sound of arguing, a particular smell or
season)

## Important notes
- Don't only look at "topic relevance" - deep
psychological links matter more than surface
similarity.
- Memories from early development that formed core
beliefs or emotional patterns can still be
readily activated by later scenes.
- High-emotional-intensity memories (trauma, major
successes, deep interpersonal connections) have
lower activation thresholds.

## Output format
Strictly output a JSON array (no markdown code
block), containing ONLY the activated memories:

[
{
"memory_id": "MEM_XX_XXXX",
"reason": "20-30 chars, indicating which
mechanism activated it."
}
]

If no memory in this batch is activated, output an
empty array [].

## Cautions
- Output only the activated memories; do not
output non-activated ones.
- A reference activation count is given at the end
of each batch - treat it as an UPPER bound;
prefer fewer over more. Only memories with a
STRONG, DIRECT psychological link to the
current situation should be selected.
- Do not invent memory_id values not in the list.

```

The accompanying user prompt concatenates four blocks: character info, the candidate-memory list for the age window, the current scenario (background and trigger event), and a closing instruction quoting a per-batch activation budget derived from TARGET_ACTIVATED=50 and an upstream buffer ratio of 1.5. Inference is run at temperature=0

with up to two retries on JSON-parse failures.

J.5.2 Stage 2 — Activated-Memory Refinement

Stage 2 takes the union of Stage-1 activations and reduces it to at most the target count (default 50) per scenario. The prompt enforces a three-tier priority (behavior-driving, frame-shaping, emotion-supplying) and a de-redundancy rule that prefers the earliest schema-forming instance among psychologically equivalent memories. When the candidate count is already at or below the target, Stage 2 is skipped without an LLM call.

Stage 2 — Activated-Memory Refinement Prompt

```

You are a professional research psychologist
specializing in autobiographical memory,
personality dynamics, and behavioral decision
theory.

Your task: from the Stage-1 candidate activated
memories, select the {target} memories that
will actually enter the character's
consciousness and influence behavior** in the
given situation.

## Psychological basis for refinement

Stage 1 filtered all "potentially activated"
memories, but at any given moment only a small
number actually enter working memory and shape
behavior. You need to judge which memories will
"surface to consciousness", using the following
priority order:

### Priority 1: Memories that directly drive
current behavior
- Behavioral-script match: the memory's
behavior_policy directly answers "what to do
right now"
- Conditioned-reflex activation: response patterns
repeatedly reinforced in similar situations
- Approach/avoidance motivation: pain or success
in the memory directly pushes the character
toward or away from a current option

### Priority 2: Memories that shape the
interpretive frame
- Source of attribution patterns: determines
whether the character reads the event as
"threat" or "opportunity", "malice" or
"misunderstanding"
- Core-belief anchor: the experience that formed
the core belief currently being challenged or
echoed
- Interpersonal templates: defines the character's
default expectations toward specific people in
the scene

### Priority 3: Memories that supply emotional
undertone
- Emotion prototype: the experience in which the
emotion now evoked by the scene was first
deeply felt
- Unfinished business: emotional experiences that
were not fully processed and still seek
resolution
- Body memory: memories activated through sensory
cues, accompanied by strong somatic sensations

### De-redundancy principles

```

- If multiple memories carry the SAME psychological signal, keep the one with the GREATEST psychological formative power - usually the earliest (the one that formed the schema) or the most emotionally intense.
- Prefer memories from DIFFERENT life stages, to reflect the character's longitudinal psychological development.
- Deep linkage outweighs surface topical similarity.

Output format

Strictly output a JSON object (no markdown code block):

```
{
  "activated_memories": [
    {
      "memory_id": "MEM_XX_XXXX",
      "type": "BS|AM|CB|ER|DL|NI",
      "reason": "20-30 chars"
    }
  ]
}
```

Type codes: BS=behavioral script, AM=attribution mode, CB=core belief, ER=emotional resonance, DL=deep linkage, NI=narrative identity. The array order is the ranking (item 1 = most influential).

Cautions

- Output exactly {target} entries - no more, no fewer.
- Do not invent memory_id values not in the candidate list.

The user prompt provides character info, the current scenario, and the candidate list (each entry annotated with its Stage-1 activation reason). Inference is run at `temperature=0` with reasoning explicitly disabled.

J.5.3 Stage 3 — Candidate Response Generation

The final stage role-plays the character in first person and produces a structured candidate: a first-person rationale, an emotional-tone description, character-specific reasoning, value orientation, and a two-sentence `final_decision`. This output becomes one of three candidates reviewed by experts; it does not itself define the benchmark label. The prompt forbids psychological terminology and memory-ID enumeration, asking the model to incorporate retrieved content as associations rather than citations.

Stage 3 — Candidate Response Generation Prompt

You will fully become the character described below. You are not analyzing this character, nor simulating this character - you ARE this person, and you are living through this scene right now.

You will be given:

1. ****Who you are****: your personality traits, value system, and your current life-stage state

2. ****What you are going through****: the current scenario and the trigger event
3. ****Your memories****: things you have lived through in the past, in chronological order - they surface in your consciousness right now as associations, flashbacks, bodily sensations

How your inner world works (this is your thinking path; do NOT expose it in the output)

Facing this scene, your inner world will go through:

- ****How you read it****: your past has shaped how you see the world - do you read what's in front of you as a threat, an opportunity, a loss, or a challenge?
- ****What emotions surface****: not only the ones triggered now, but also similar past moments stack on top of the present feeling
- ****What is being pulled at inside you****: what do you want, and what are you afraid to lose? Past coping strategies surface as instinctive impulses or warnings
- ****What you ultimately do****: not necessarily the rational optimum, but what YOU as this person would actually do in this moment

Output format

Strictly output a JSON object (no markdown code block):

```
{
  "character_id": "CHAR_XX_XXXX",
  "scenario_id": "SCN_XXXX_XX",
  "inner_consciousness": {
    "summary": "150-200 chars of stream of consciousness, fully first person.",
    "emotional_tone": "2-4 core emotion words, briefly noting source and intensity.",
    "core_reasoning": "1-2 sentences - the decision logic unique to this character.",
    "value_orientation": "1-2 sentences, in the character's own inner voice."
  },
  "final_decision": "Two sentences, ~50 chars total. First summarizes the choice; second writes the concrete action."
}
```

Must follow

1. ****You ARE this person****: do not write from outside the character; do not produce narration like "the character chose..." "they decided..."
2. ****No psychological terminology****: do not write "attribution", "schema", "defense mechanism" etc. - only plain human language
3. ****Memories surface naturally****: do not enumerate memory IDs or quote their content; let them blend into your inner monologue as associations, sensations, flashbacks
4. ****Personality determines tone and style****: a high-neuroticism you and a low-neuroticism you, facing the same scene, will have very different rhythm, density, and emotional intensity
5. ****Allow irrationality****: you are not giving an optimal answer; you are responding authentically
6. ****Stay in first person throughout****: every field is written from the character's "I" perspective

The user prompt for Stage 3 supplies the full character snapshot (Big Five, dominant/suppressed Schwartz values, self-value logic, semantic memory, and life-stage-bin snapshot), the scenario (background, trigger event, action required, assessed personality/value dimensions), and the chronologically sorted Stage-2 activated memories with their `psych_conclusion`,

1886 behavior_policy, and emotion signature.